

Air Quality 101

A Primer on Air Quality

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AQ 101: Presentation Content

- ❖ **Definitions** of air pollution and air quality
- ❖ **History** of air pollution
- ❖ **Regulation** of air pollution in the U.S.
- ❖ **Common pollutants** in the air
- ❖ **Sources** of air pollution
- ❖ **Air Monitoring: Air Sensors** - Newer ways to collect air quality measurements
- ❖ **Source to Receptor Paradigm: Emissions, Exposures, Health Effects**
- ❖ **Health effects** of air pollutants
- ❖ **Common questions**
 - ❖ Why and when air pollution can be higher and lower?
 - ❖ How to tell what the air quality is and where to find information?
 - ❖ How are air pollutants are measured and where to find the data?
- ❖ **Air Monitoring:** Air Sensors vs. EPA's Federal Reference Monitors (FRMs) vs. Federal Equivalent Monitors (FEMs)
- ❖ **Additional resources**
- ❖ **References**



June 26, 2023 June 27, 2023

AQ 101: Is this Air Clean?



Good visibility suggests lower pollution levels!

AQ:101 Is this Air Clean?



Los Angeles, California



Shanghai, China (at Sunset)



Chicago, Illinois



New York City, New York



Cairo, Egypt

**These examples
show poor air
quality events!**

AQ 101: What is Air Pollution and Air Quality?

Air Pollution: A contaminant or mixture of contaminants in the air that can harm humans, plants, animals, and the environment.



Air Quality is a word used to describe how much pollution is in the air.

Good air quality = less pollution



Poor air quality = more pollution



AQ101: History of Air Pollution



Why do we care about air pollution?

**Let's look at the air over
50 years ago.....**



Source: <https://www.chicagotribune.com/2015/06/05/for-much-of-its-history-chicago-covered-by-smoke-soot/>, Downtown Chicago, February 1, 1957.

AQ 101: History of Smog

Where did the word “smog” come from?

The word “smog” was created in the early 1900’s to describe the environment of sooty fog in England.

Smoke and fog = Smog

1300

In England, King Edward II banned coal burning in London while parliament was in session and announced that *“whosoever shall be found guilty of burning coal shall suffer the loss of his head”*.

1943

The first known episodes of smog in the U.S. occurred in Los Angeles. Visibility was only three blocks and people suffered from itchy eyes, breathing problems, nausea, and vomiting.



Protests at Pasadena City Hall on November 9, 1954, following 15 days of smog in October.

AQ 101: Examples of Air Pollution in the Past

Some causes of poor air quality....

Poor or no air pollution controls



Inefficient fuel burning



Some results of poor air quality....



Air quality in New York City on Thanksgiving Day in 1966



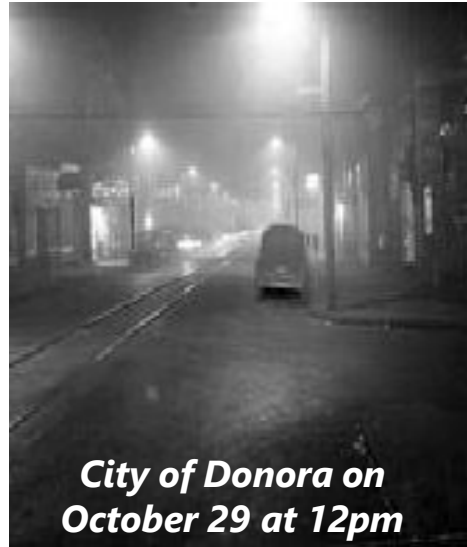
Smog over Phoenix, Arizona in 1973

AQ 101: A Few Important Smog Episodes

Donora, Pennsylvania

A smog episode occurred on October 26-31, 1948, due to toxic emissions released from a local steel plant.

20 people died and 6,000 people suffered illness...out of population of 14,000.



London, England

A smog episode occurred on December 4-9, 1952, due to high sulfur dioxide emissions.

Approximately 4,000 people died.



These and many other air pollution episodes helped paved the way for the regulations we have today to help protect human health and the environment.

AQ 101: Air Pollution in Chicago



Pilsen- where I'm pointing is supposed to be downtown



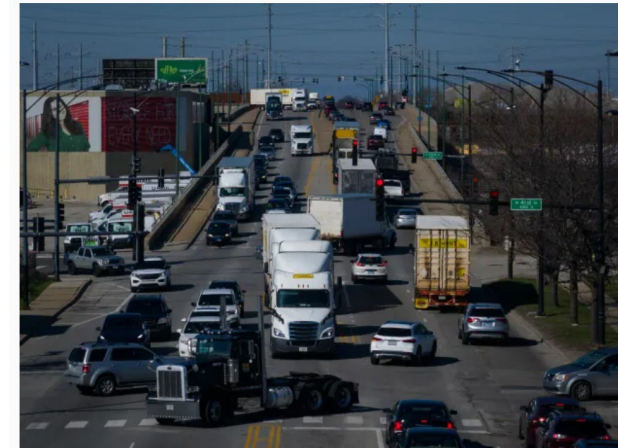
7:55 PM · Jun 27, 2023



My normal view from the office lunch room compared to today's view (medical district)



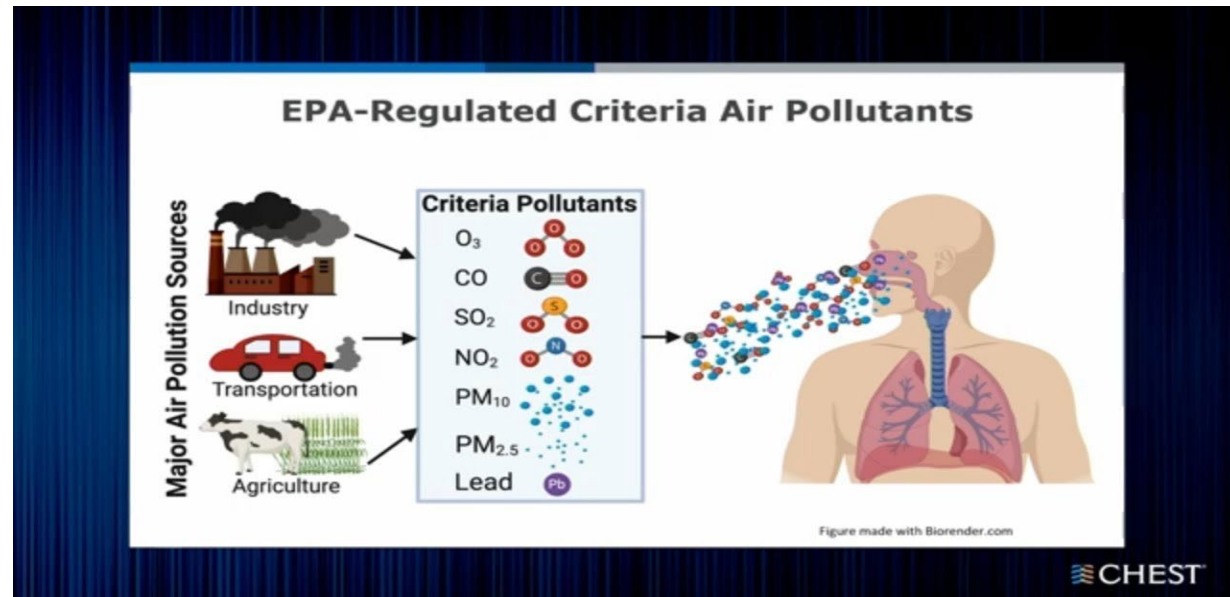
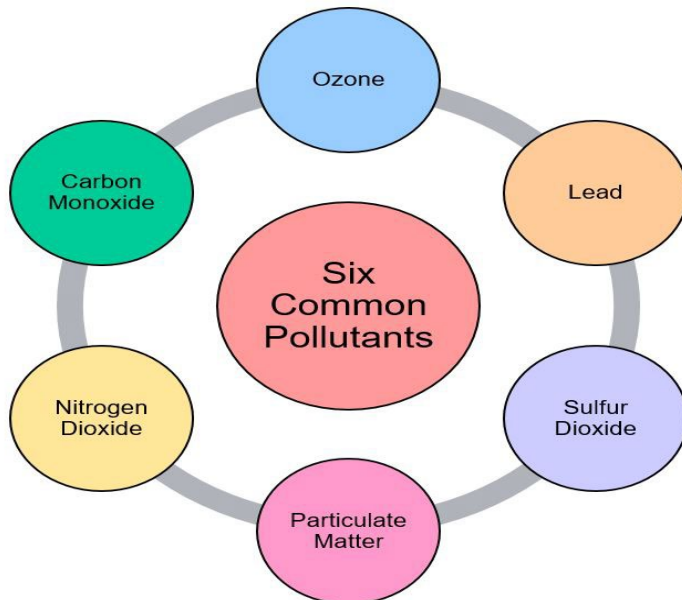
10:21 PM · Jun 27, 2023



Photos from Chicago Tribune,
Block Club Chicago, and
Instagram

AQ 101: Air Pollution Regulation in the U.S.

- ❖ The **Clean Air Act** was signed into law in 1970 with major changes in 1990: Find out more about the Clean Air Act here: <https://www.epa.gov/clean-air-act-overview>
- ❖ **EPA regulates air pollution from 3 different sources in general:** 1) Stationary sources (i.e., industrial facilities); 2) Mobile sources (light/heavy duty vehicles, airplanes, railways); and 3) Area sources (gas stations, dry cleaners, etc.)
- ❖ Under the Act, the U.S. Environmental Protection Agency (EPA) is required to set **health-based National Ambient Air Quality Standards (NAAQS)** for **the 6 criteria pollutants**



Source: ACCP, Chest, April 22, 2025, Presentation

AQ 101: National Ambient Air Quality Standards (NAAQS)

Pollutant [links to historical tables of NAAQS reviews]		Primary/ Secondary	Averaging Time	Level	Form
Carbon Monoxide (CO)		primary	8 hours	9 ppm	Not to be exceeded more than once per year
			1 hour	35 ppm	
Lead (Pb)		primary and secondary	Rolling 3 month average	0.15 µg/m ³ (1)	Not to be exceeded
Nitrogen Dioxide (NO ₂)		primary	1 hour	100 ppb	98th percentile of 1-hour daily maximum concentrations, averaged over 3 years
		primary and secondary	1 year	53 ppb (2)	Annual Mean
Ozone (O ₃)		primary and secondary	8 hours	0.070 ppm (3)	Annual fourth-highest daily maximum 8-hour concentration, averaged over 3 years
Particle Pollution (PM)	PM _{2.5}	primary	1 year	9.0 µg/m ³	annual mean, averaged over 3 years
		secondary	1 year	15.0 µg/m ³	annual mean, averaged over 3 years
		primary and secondary	24 hours	35 µg/m ³	98th percentile, averaged over 3 years
	PM ₁₀	primary and secondary	24 hours	150 µg/m ³	Not to be exceeded more than once per year on average over 3 years
Sulfur Dioxide (SO ₂)		primary	1 hour	75 ppb (4)	99th percentile of 1-hour daily maximum concentrations, averaged over 3 years
		secondary	1 year	10 ppb	annual mean, averaged over 3 years

- There are 2 types of **NAAQS**:
 - ❖ **Primary Standard** – protects public health including "sensitive" populations such as asthmatics, children, and the elderly.
 - ❖ **Secondary Standard** – protects public welfare including protection against decreased visibility and damage to animals, crops, vegetation, and buildings.

Different **averaging times** and **levels** are set for the primary and secondary NAAQS for each criteria pollutant.

For more details:

<https://www.epa.gov/criteria-air-pollutants/naaqs-table>



AQ 101: Common Pollutants in the Air



NAAQS Criteria Pollutants

❖ Particulate Matter (PM)



❖ Gases including...

- ❖ Ozone (O₃)
- ❖ Carbon monoxide (CO)
- ❖ Nitrogen dioxide (NO₂)
- ❖ Sulfur dioxide (SO₂)

More information can be found here:
<https://www.epa.gov/criteria-air-pollutants>

Hazardous Air Pollutants (HAPs)

HAPS are pollutants that are known or suspected to cause cancer or other serious health effects. There are 187 HAPs!

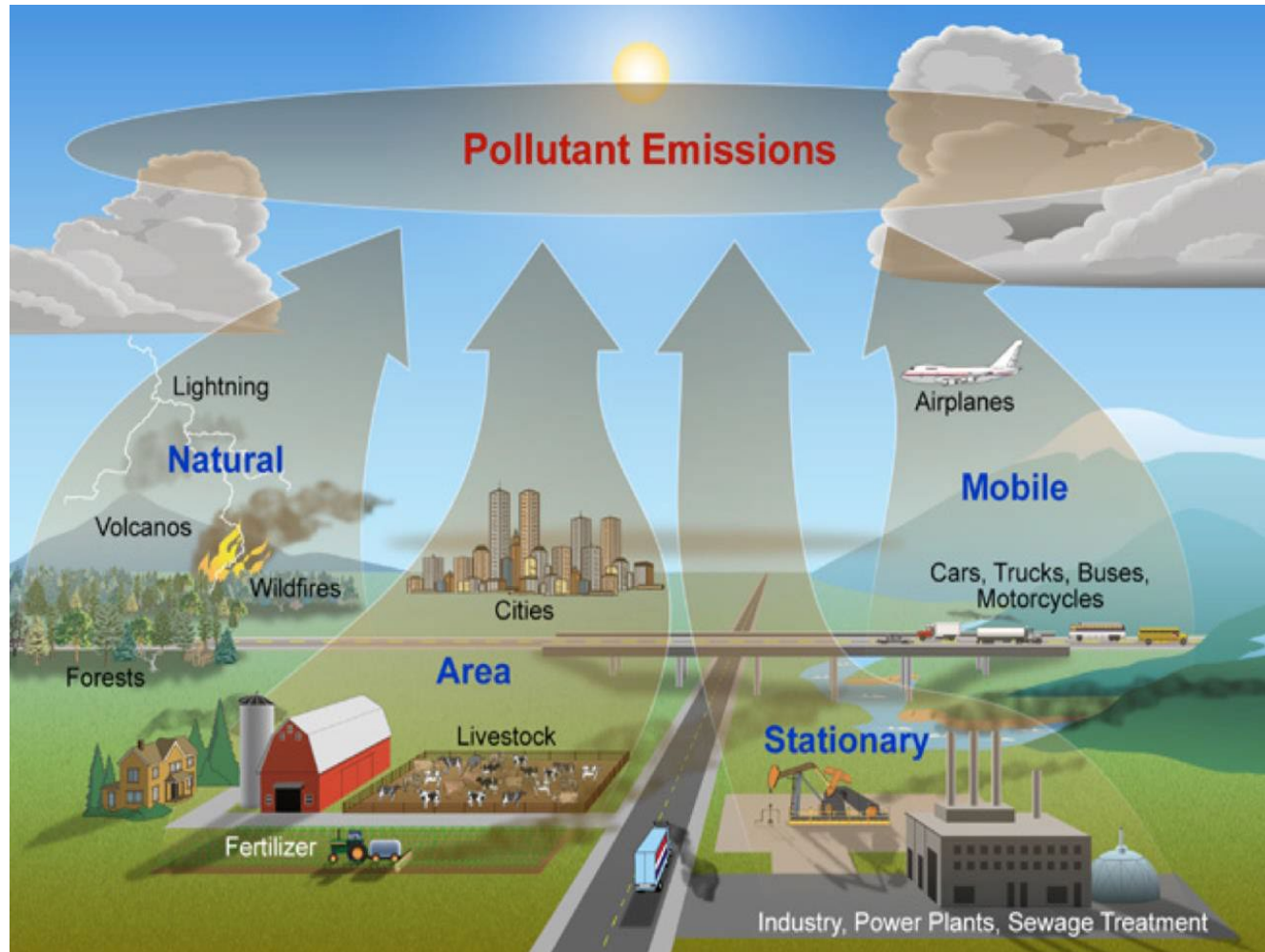
Examples of **HAPs** include...

- ❖ Asbestos
- ❖ Benzene
- ❖ Toluene
- ❖ Mercury

HAPs are regulated under a different part of the Clean Air Act. More information can be found here: <https://www.epa.gov/haps>

- There are two physical/chemical state of air pollutants in ambient air: 1) Particulates (Dust) in Air: and 2) Gaseous Pollutants in ambient air

AQ 101: Where Does Pollution Come From?



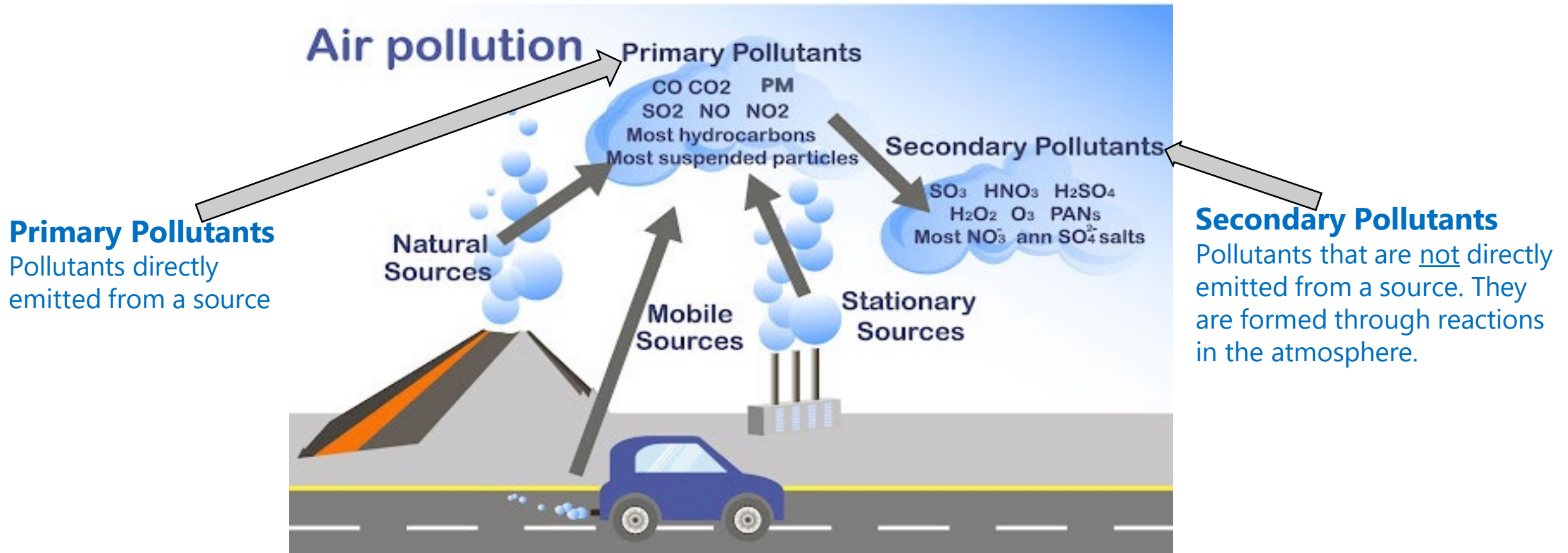
Natural sources such as...

- ❖ Volcanoes
- ❖ Wildfires/forest fires
- ❖ Lightning
- ❖ Wind and dust storms
- ❖ Animals
- ❖ Soil dust

Man-made sources such as...

- ❖ Cars, trucks, buses, motorcycles, airplanes, ships
- ❖ Fossil fuel burning (coal, oil)
- ❖ Industries, power plants, factories
- ❖ Meat cooking/barbequing
- ❖ Burning wood, leaves, or vegetation

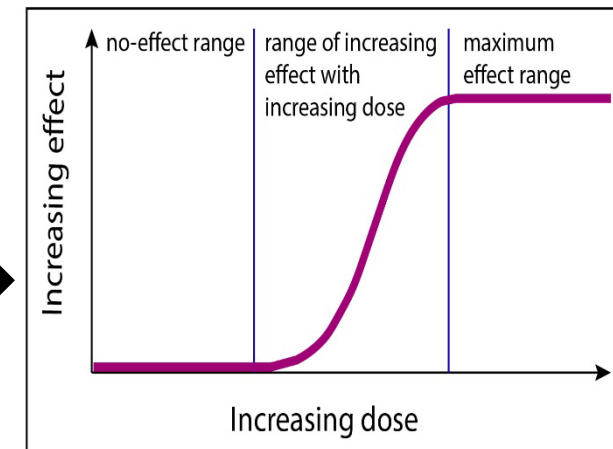
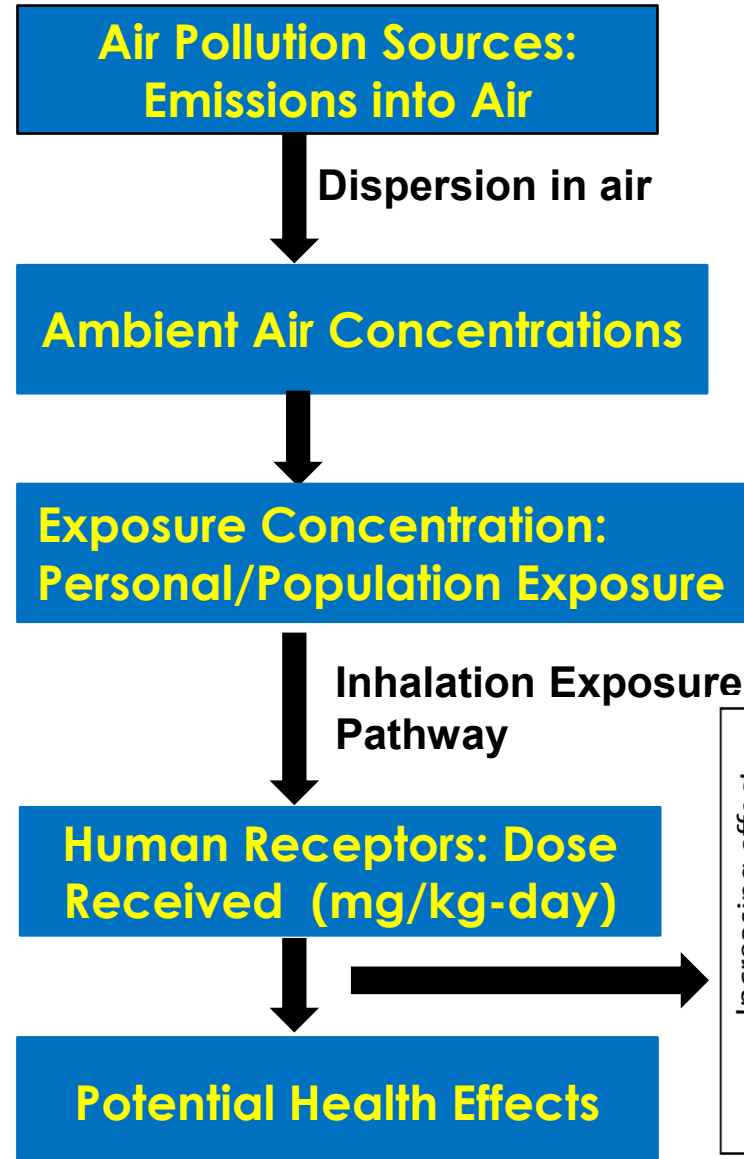
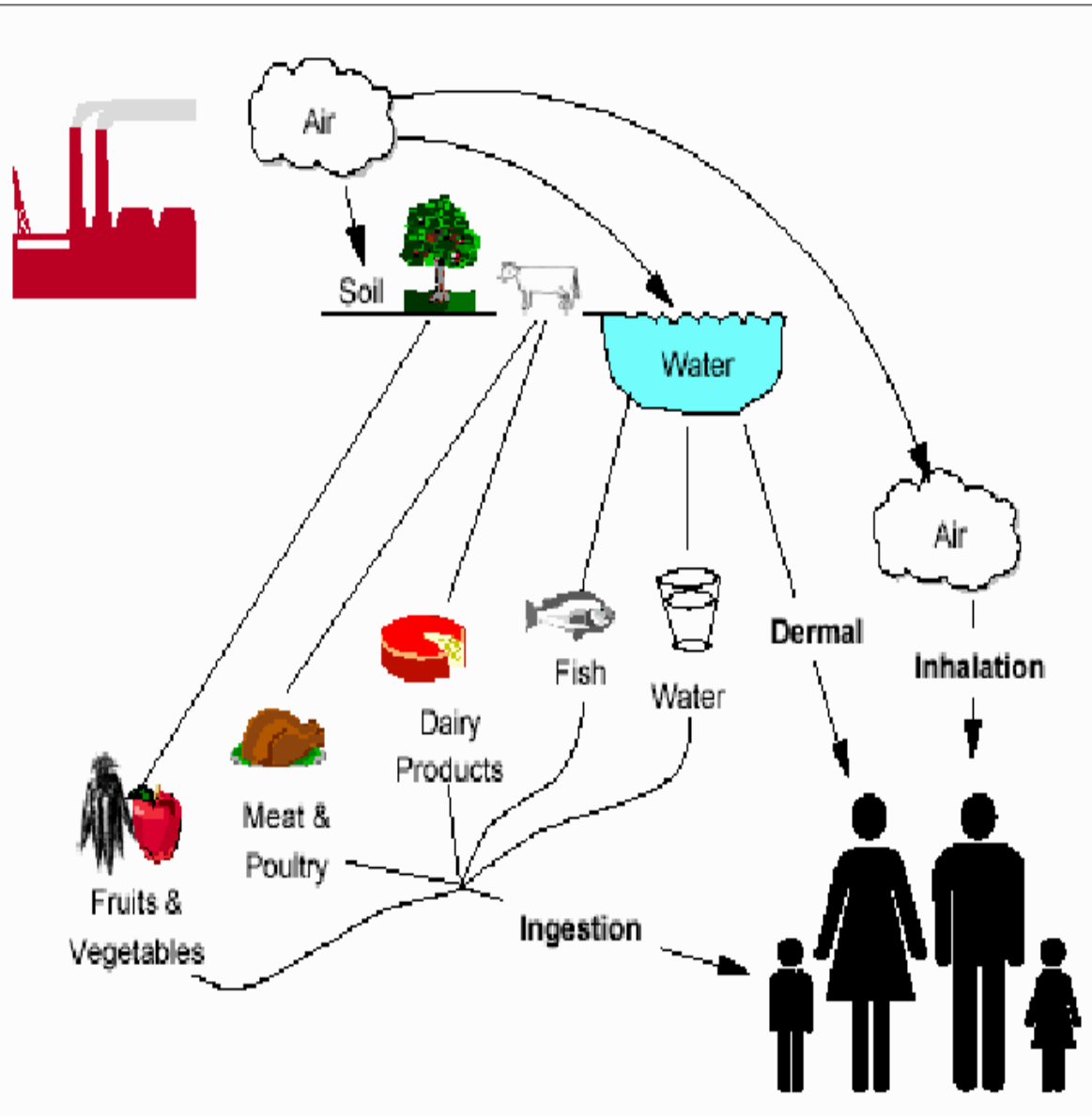
AQ 101: Primary versus Secondary Pollutants



Source: <https://www.mrgscience.com/ess-topic-63-photochemical-smog.html>

Particulate matter and **ozone** cause the most concern for health in the U.S.
Let's take a closer look at these pollutants, along with nitrogen dioxide!

AQ 101: Air Pollutants: Source to Receptor Paradigm



Health effects of air pollution

short-term effects

exacerbation
of asthma

cough, wheezing
and shortness
of breath

episodes of high air
pollution increase
respiratory and
cardiovascular hospital
admissions and mortality

long-term effects

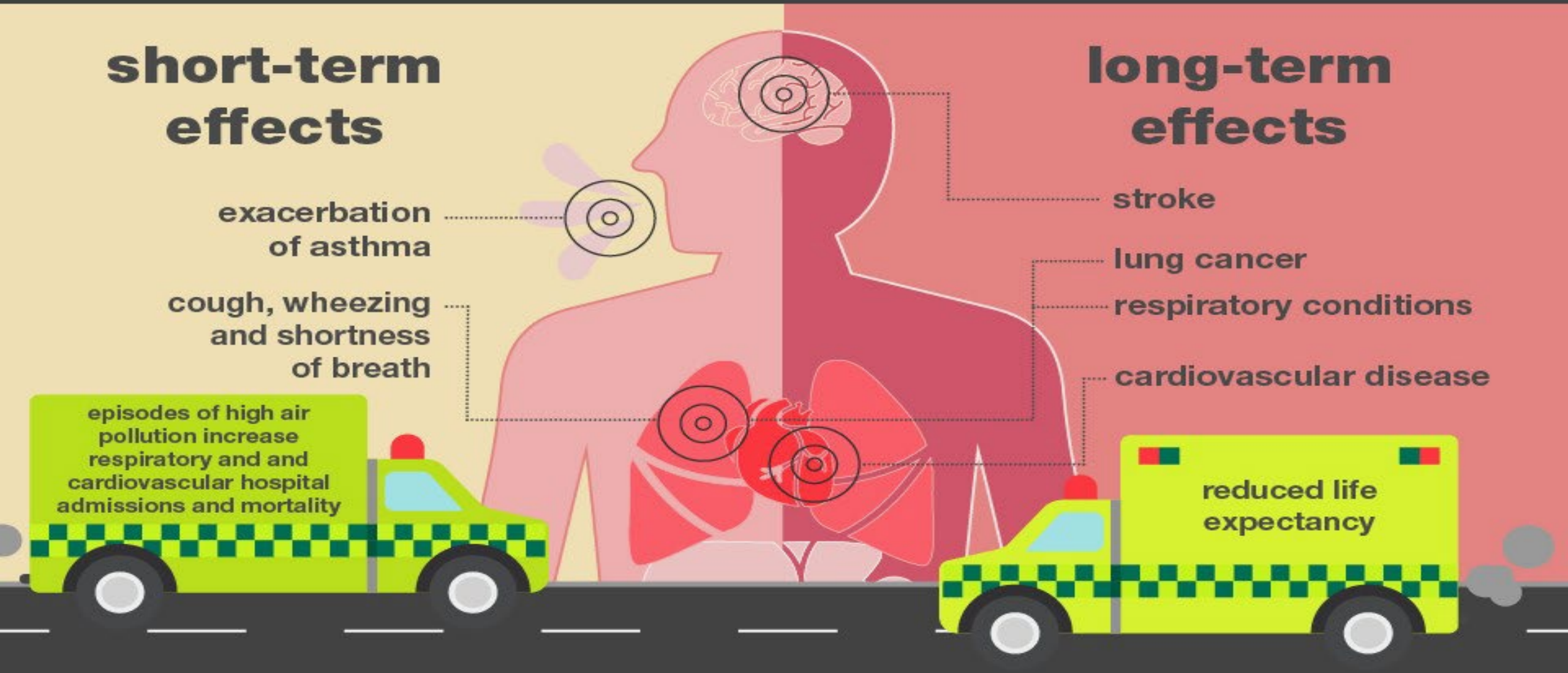
stroke

lung cancer

respiratory conditions

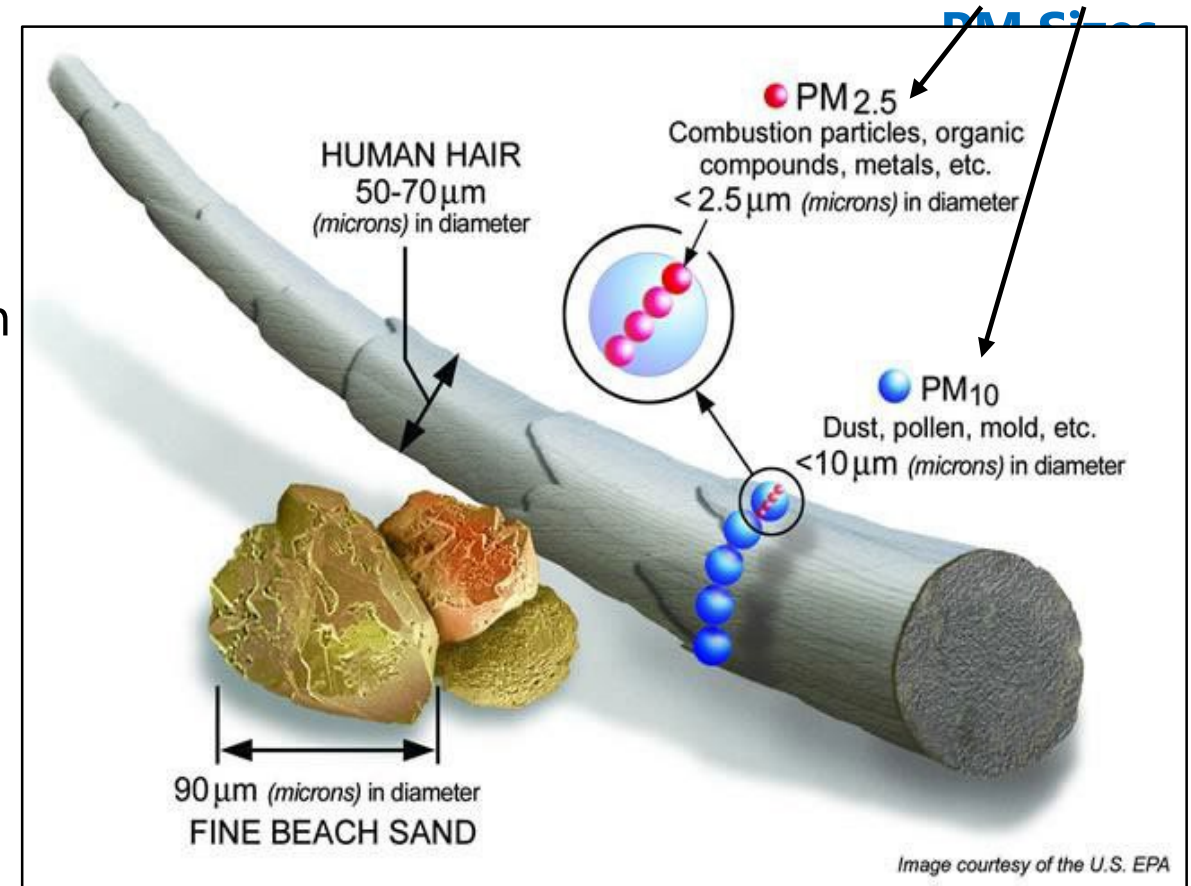
cardiovascular disease

reduced life
expectancy



AQ 101: What is Particulate Matter (PM)?

- ❖ **Particulate matter (PM)** is a mixture of solid particles and liquid droplets found in the air. **PM** can be made of dust, dirt, soot, smoke, pollen, metals, and other particles
- ❖ Some particles can be seen by the eye others can
 - ❖ only be seen using a microscope
- ❖ **PM** is both a **primary pollutant** and **secondary pollutant**
- ❖ Common **sources of PM** include...
 - ❖ Mobile sources (cars, trucks, buses, others)
 - ❖ Industry, power plants, factories
 - ❖ Wildfires/wood burning
 - ❖ Windblown dust
 - ❖ Unpaved roads

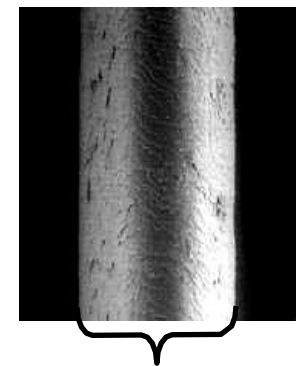


Recall: PM is one of the criteria pollutants regulated under the NAAQS!

AQ 10: Particle Size is Very Important! -- Particle size determines where particles will deposit in the respiratory system once inhaled

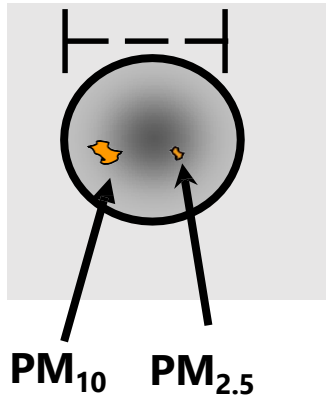
- ❖ Two common **particulate matter (PM)** size fractions include
 - ❖ **PM_{2.5}** – also called 'fine PM', have diameters of 2.5 micrometers or smaller
 - ❖ **PM₁₀** – also called 'coarse PM', have diameters of 10 micrometers or smaller

Some particles less than 10 micrometers (μm) in diameter can get deep into your lungs and some may even get into your bloodstream.
PM_{2.5} has the most risk to health!

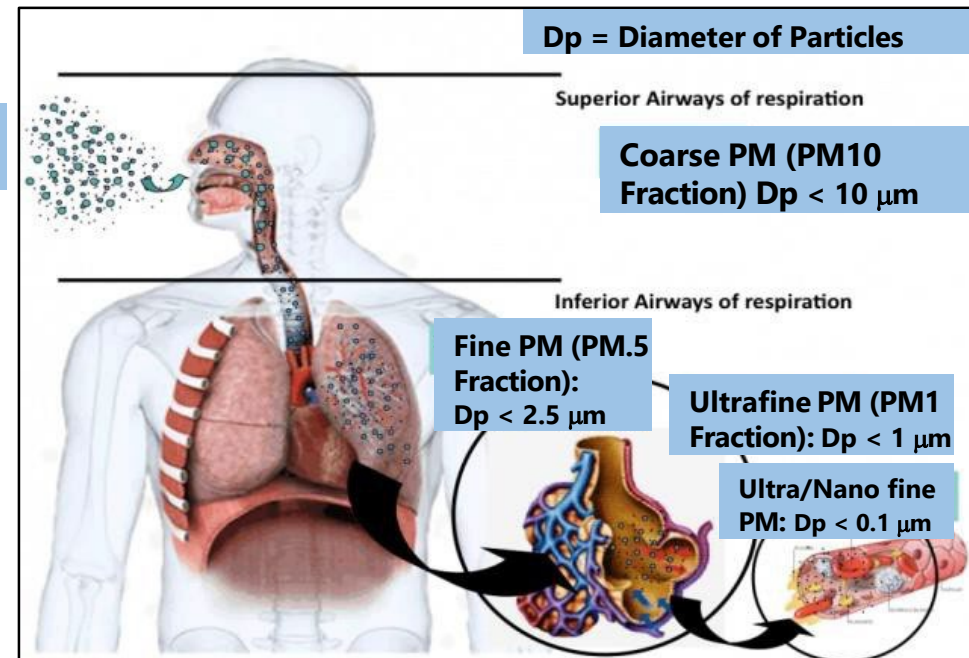


Human Hair
(70 micrometers)

Hair cross section
(70 micrometers)



Inhalable PM
Fraction $D_p < 100 \mu\text{m}$



AQ 101: Particulate Matter (PM) – Health Effects

- ❖ **PM** can affect both your **lungs** and **heart**
- ❖ **Exposure** to **PM** can cause
 - ❖ Nonfatal heart attacks
 - ❖ Irregular heartbeat
 - ❖ Premature death in people with heart and lung disease
 - ❖ Aggravated asthma
 - ❖ Decreased lung function
 - ❖ Increased respiratory systems such as irritation of airways, coughing, difficulty breathing

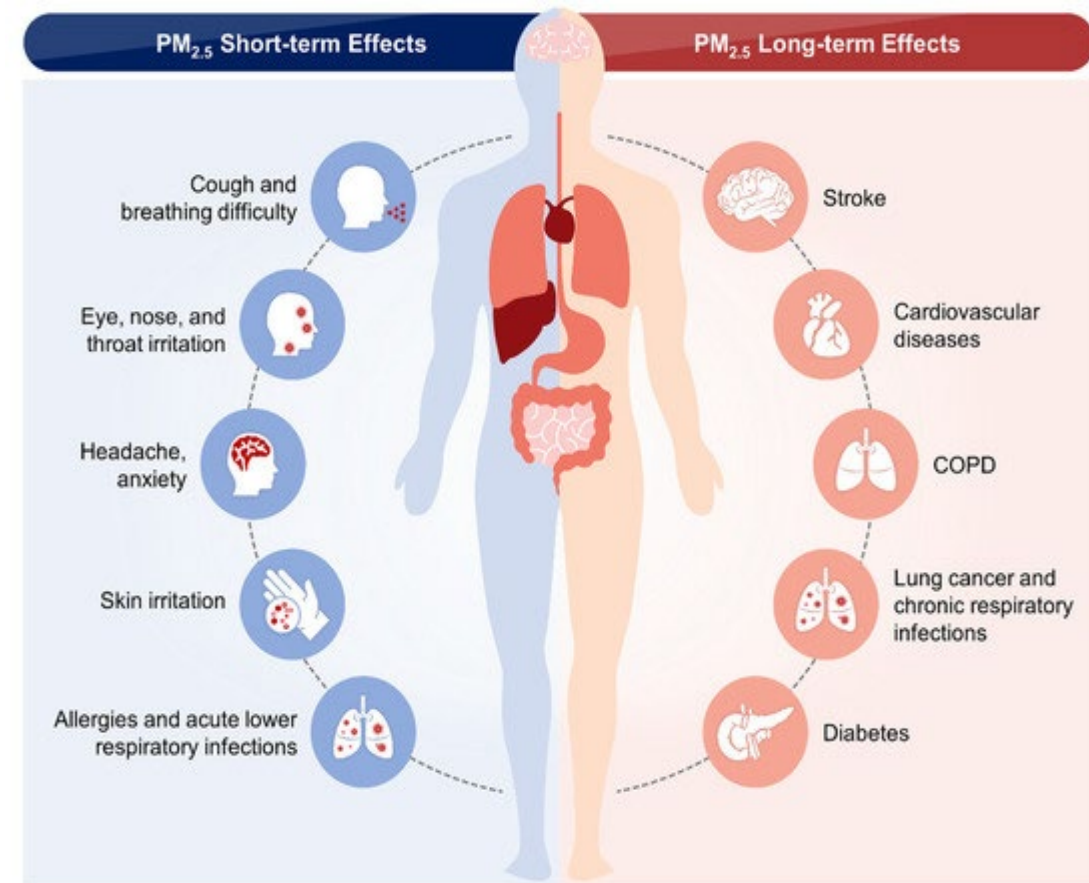
People with heart or lung disease, children, and older adults are **most likely** to be impacted by **PM**.

Inside Airway

Normal



Red and Swollen

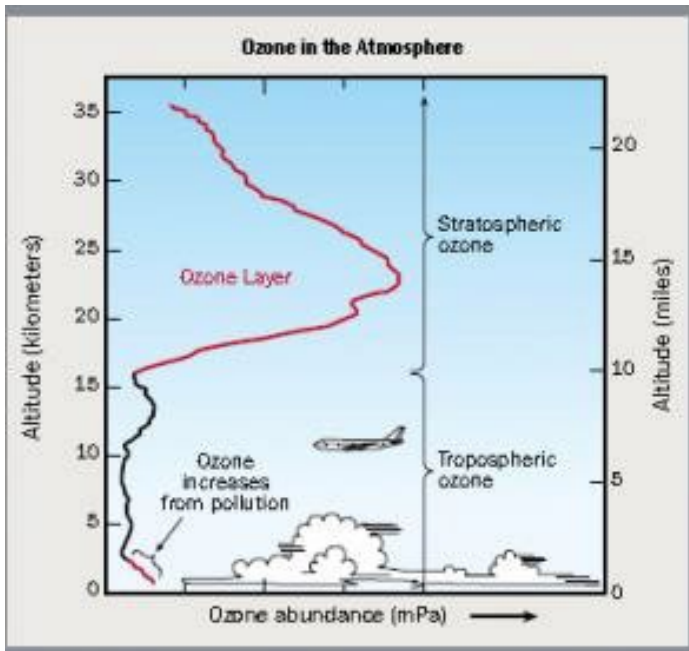


Source: Basith S, et al.. The Impact of Fine Particulate Matter 2.5 on the Cardiovascular System: A Review of the Invisible Killer. Nanomaterials (Basel). 2022 Aug 2;12(15):2656. doi: 10.3390/nano12152656.

AQ 101: What is Ozone (O₃)?

Good ozone

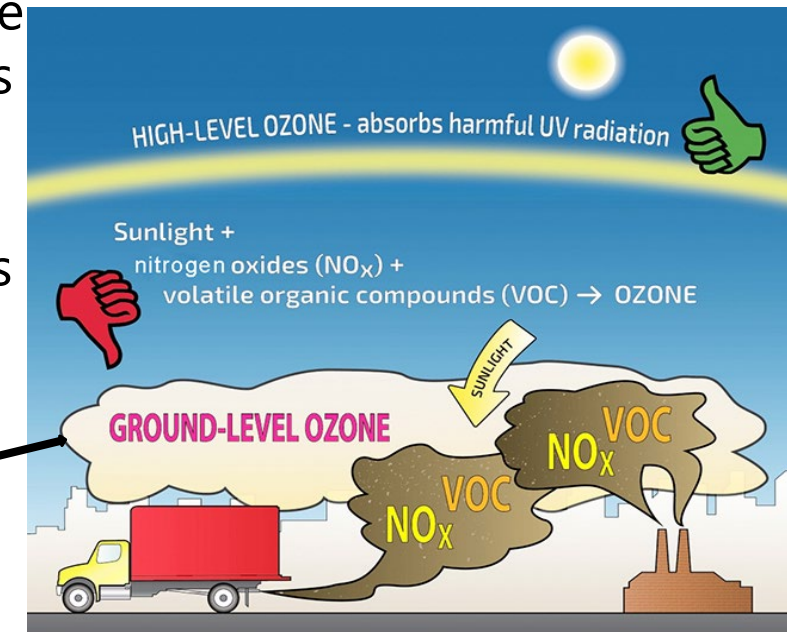
- ❖ Also called **stratospheric ozone**
- ❖ Found in the upper atmosphere
- ❖ Forms a protective layer that shields us from the sun's harmful ultraviolet (UV) rays



Bad ozone

- ❖ Also called **ground-level ozone**
- ❖ Found in lower atmosphere (near ground level)
- ❖ Secondary pollutant only
- ❖ Major sources that emit pollutants which form ozone include
 - ❖ Industrial facilities
 - ❖ Electric utilities
 - ❖ Motor vehicles
 - ❖ Chemical solvents

How ground-level ozone forms



Recall: Ozone is one of the criteria pollutants regulated under the NAAQS!

AQ 101: Ozone – Health Effects

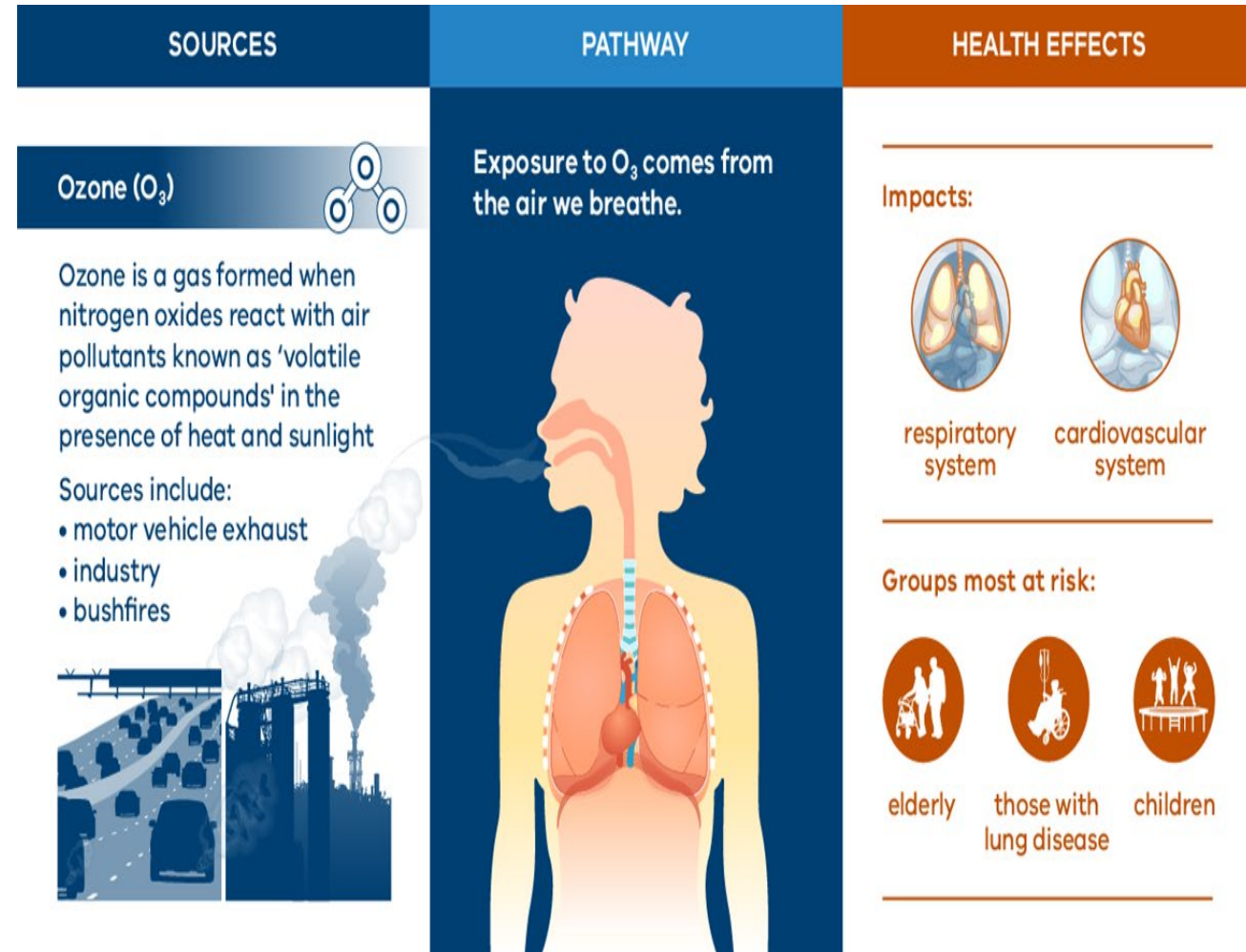
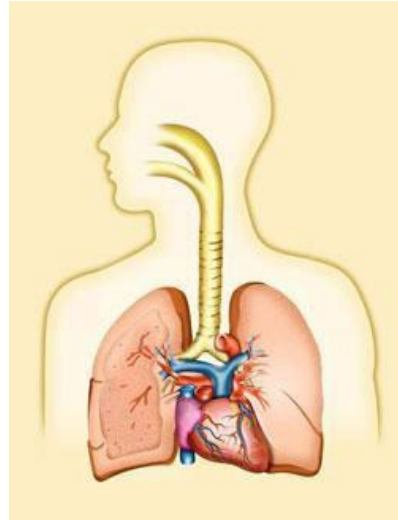
Ozone in the air we breathe can harm our health.

❖ **Breathing ozone can cause**

- ❖ Chest pain
- ❖ Coughing
- ❖ Throat irritation
- ❖ Inflammation of airways
- ❖ Lung function decrements

❖ **Ozone can worsen**

- ❖ Asthma
- ❖ Bronchitis
- ❖ Emphysema

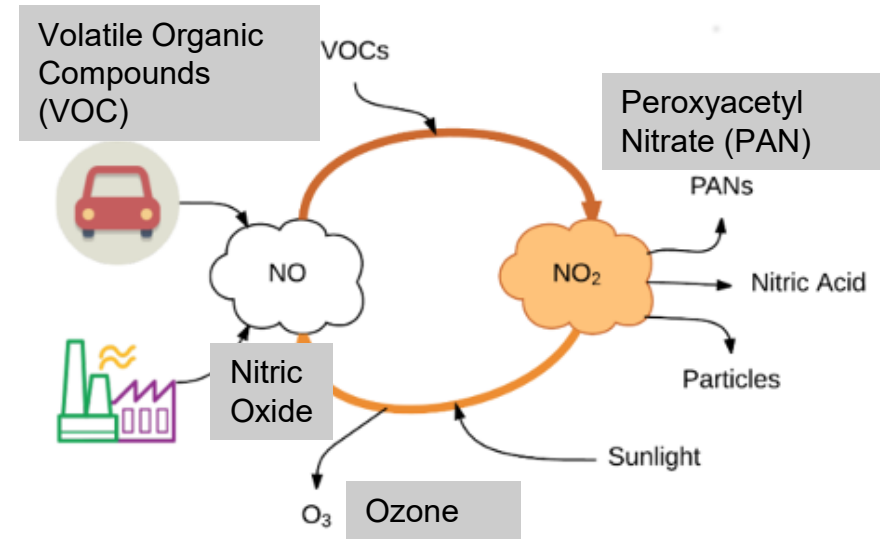


Source: <https://pirg.org/colorado/foundation/articles/raacs-sips-naaqs-oh-my-heres-what-the-acronyms-mean-and-why-theyre-critical-for-reducing-ozone-pollution/>, October 25, 2022.

People **most** at risk include people with asthma, children, older adults, and people who are active or work outdoors.

AQ 101: What is Nitrogen Dioxide (NO₂)?

- ❖ Nitrogen Dioxide (NO₂) is one of a group of highly reactive gases known as oxides of nitrogen or nitrogen oxides ($\text{NO} + \text{NO}_2 = \text{NO}_x$). Other nitrogen oxides include nitrous acid and nitric acid. **NO₂ is used as the indicator for the larger group of nitrogen oxides.**
- ❖ **NO₂ primarily is released into air from the burning of fuel.** NO₂ forms from emissions from cars, trucks and buses, power plants, and off-road equipment.
- ❖ Although NO₂ can be directly emitted from combustion sources, much of the NO₂ in the ambient air is formed in the atmosphere through reactions between nitric oxide (NO) and other air pollutants that require the presence of sunlight (photochemical reactions). **NO₂ contributes to formation of several other air pollutants, including ozone (O₃), nitric acid (HNO₃), and nitrate (NO₃⁻) -containing particles** that also form through photochemical reactions.



Source: <https://aqicn.org/faq/2017-01-10/nitrogen-dioxide-no2-in-our-atmosphere/> , January 10, 2017.

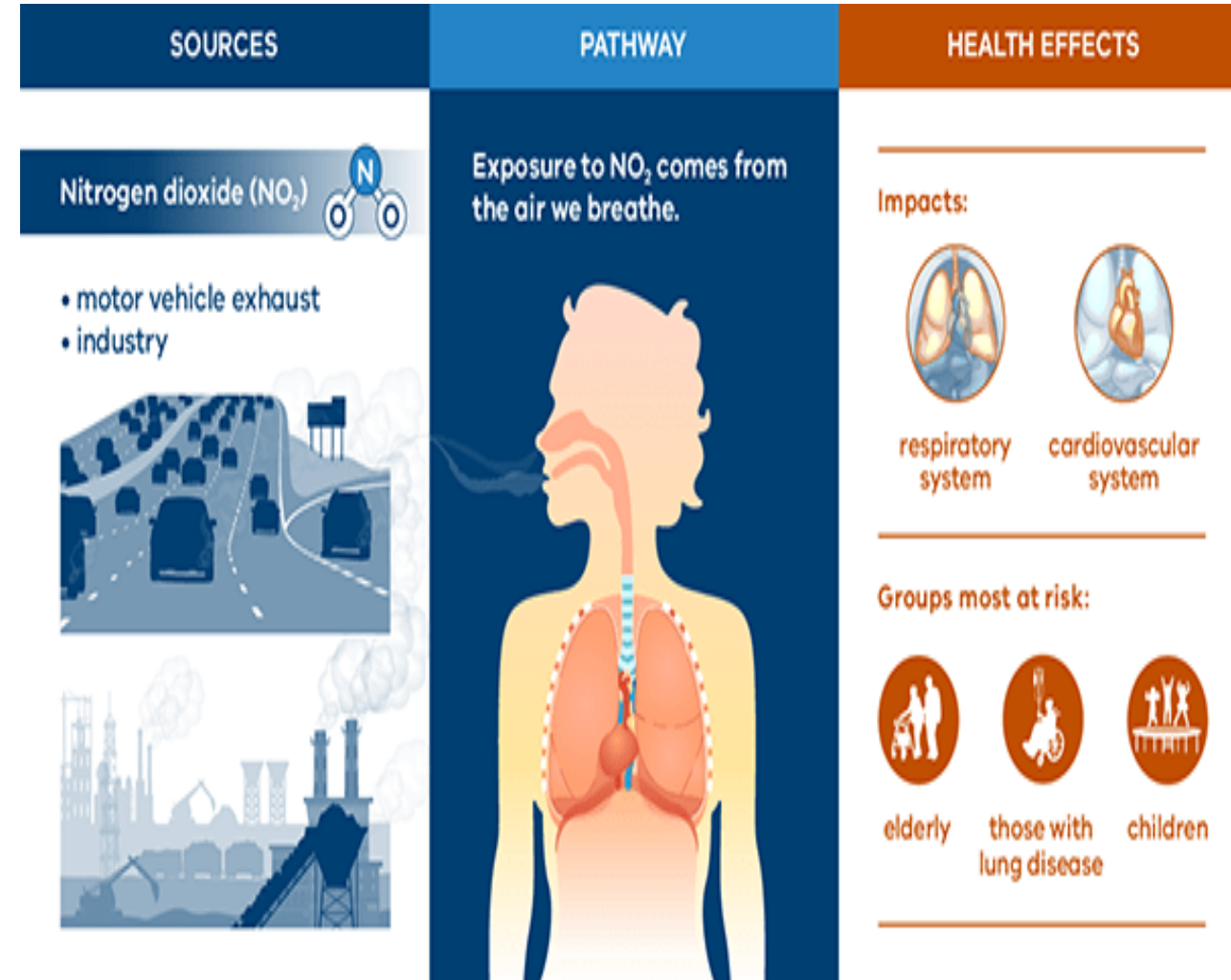
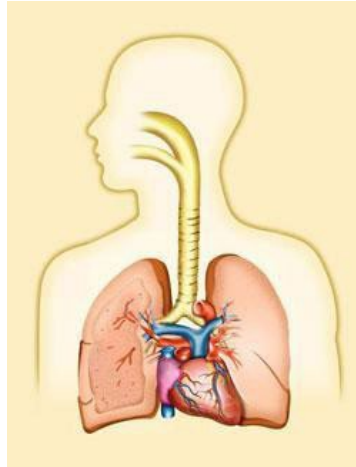
Nitrogen dioxide is considered a primary pollutant because it is directly emitted from sources, but it also acts as a precursor to other pollutants, making it a factor in secondary pollution as well.

Recall: Nitrogen Dioxide is one of the criteria pollutants regulated under the NAAQS!

AQ 101: Nitrogen Dioxide (NO₂) – Health Effects

Nitrogen Dioxide in the air we breathe can harm our health.

- ❖ Short-term exposures to high concentrations of NO₂ can irritate airways, aggravate respiratory diseases, particularly asthma, leading to respiratory symptoms (such as coughing, wheezing or difficulty breathing), hospital admissions and visits to emergency rooms.
- ❖ Longer exposures to elevated concentrations of NO₂ may contribute to the development of asthma and potentially increase susceptibility to respiratory infections.



People with asthma, as well as children and the elderly are generally at greater risk for the health effects of NO₂.

Source: <https://www.epa.vic.gov.au/nitrogen-dioxide>, January 29, 2025.

AQ 101: Why and When Can Air Pollution Levels be Higher or Lower?

Emissions from Sources

More emissions from sources, makes more air pollution



Weather Conditions



Sunlight

Heat from the sun can cause chemical reactions that make pollution.



Rain

Rain washes pollution out of the air.

Wind

Pollution stays around in stagnant conditions. On windy days, pollution is spread around.

Temporal Variation (changes over time)

Time of the day or week

During rush hour, pollution levels are often higher since people are traveling to work/school.

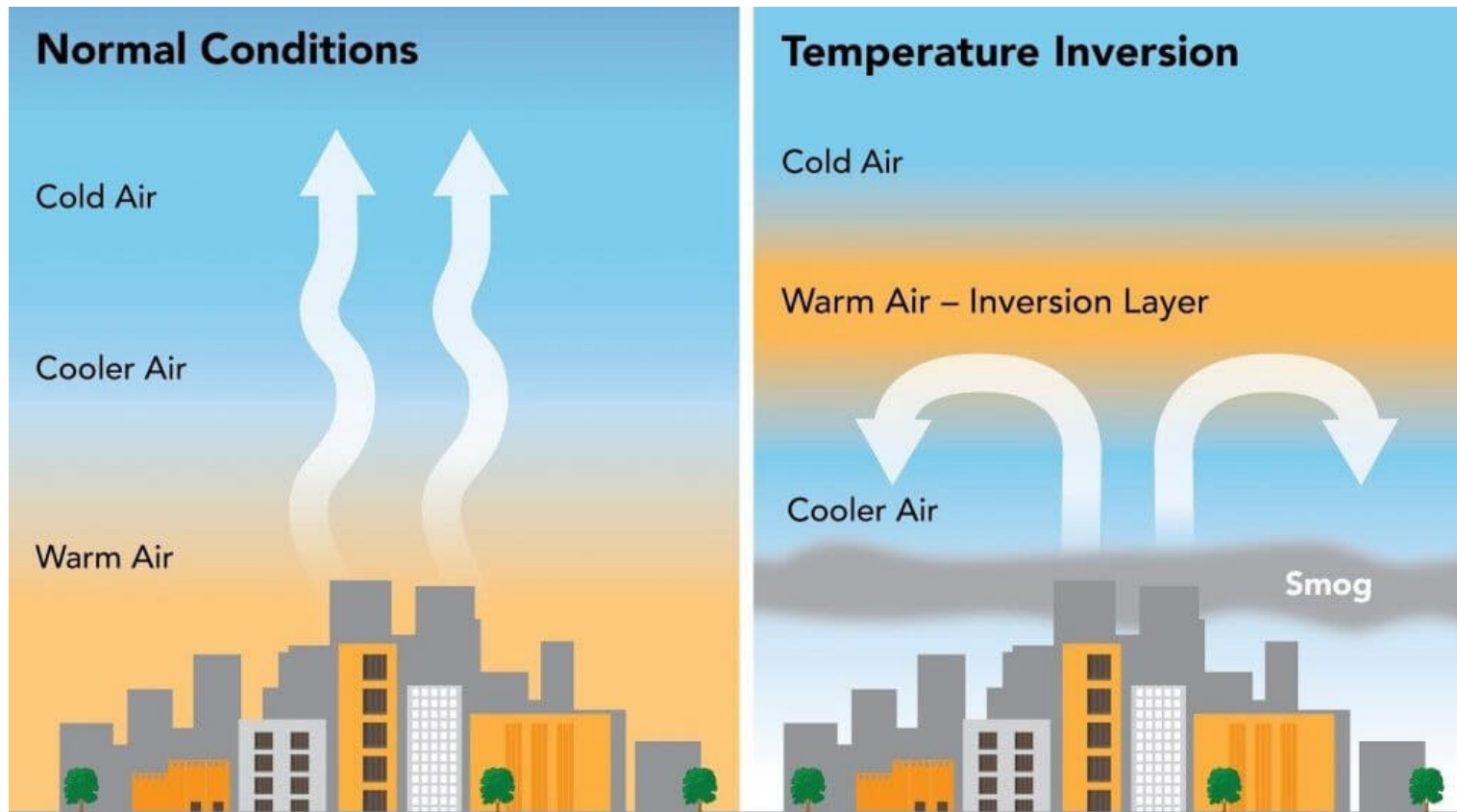
On the weekends, pollution levels are often lower since less people work/go to school.

Seasons

Levels of pollutants like ozone, created in part using sunlight, are often higher in summer and lower in winter. For other pollutants, the levels may also depend on sources. For example, burning for heat increases PM in cooler months.

AQ 101: Special Weather Condition that Causes Higher Air Pollution Levels

Temperature Inversions due to weather conditions can also increase air pollution levels.



Temperature Inversion in Downtown Los Angeles, CA
Source: <https://blog.tempest.earth/temperature-inversions-what-how-and-why/>

Normally, air temperature becomes **colder** higher above the ground

During a **temperature inversion**, a warm layer is 'sandwiched' in between cooler layers. Pollution can be trapped, causing higher pollution levels where we live!

AQ 101: How Can You Tell What the Air Quality is?

Sometimes you can **see** pollution. The sky could be hazy and brownish.



Source: <https://www.facebook.com/photo.php?fbid=5811564675597749&id=532732413481028&set=a.543325305755072>, Butler Photography, February 5, 2023.

Sometimes you can **smell** pollution. For example, smoke from wildfires.



Source: <https://abc7chicago.com/chicago-air-quality-alert-candian-wildfire-smoke/13509894/>, July 16, 2023

Other ways to check the air quality:

- ✓ AirNow website (www.airnow.gov)
- ✓ Local news for Ozone Action Days, Air Quality Index reports in radio, TV, or newspaper (usually in the weather report!)

EPA's Air Quality Index (AQI)

AQI Basics for Ozone and Particle Pollution

Daily AQI Color	Levels of Concern	Values of Index	Description of Air Quality
Green	Good	0 - 50	Air quality is satisfactory, and air pollution poses little or no risk.
Yellow	Moderate	51 - 100	Air quality is acceptable. However, there may be a risk for some people, particularly those who are unusually sensitive to air pollution.
Orange	Unhealthy for Sensitive Groups	101 -150	Members of sensitive groups may experience health effects. The general public is less likely to be affected.
Red	Unhealthy	151 - 200	Some members of the general public may experience health effects; members of sensitive groups may experience more serious health effects.
Purple	Very Unhealthy	201 - 300	Health alert: The risk of health effects is increased for everyone.
Maroon	Hazardous	301 and higher	Health warning of emergency conditions: everyone is more likely to be affected.

- ❖ **The U.S. Air Quality Index (AQI) is EPA's tool for communicating about outdoor air quality and health** (<https://www.airnow.gov/aqi/aqi-basics/>) .
- ❖ The AQI includes six color-coded categories, each corresponding to a range of index values. The higher the AQI value, the greater the level of air pollution and the greater the health concern. The color makes it easy for people to quickly determine whether air quality is reaching unhealthy levels in their communities.
- ❖ For each pollutant an AQI value of 100 generally corresponds to an ambient air concentration that equals the level of the short-term national ambient air quality standard for protection of public health. AQI values at or below 100 are generally thought of as satisfactory. When AQI values are above 100, air quality is unhealthy: at first for certain sensitive groups of people, then for everyone as AQI values get higher.

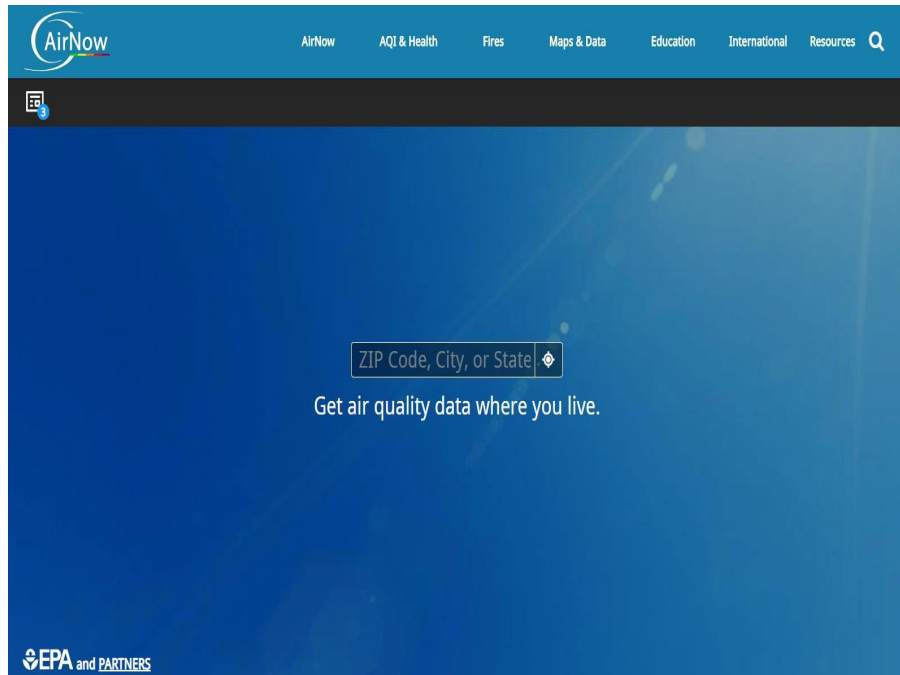
Ozone and **particulate matter** pollution cause the most concern for health in the U.S.

AQ 101: EPA's AirNow Website

AirNow is a system that provides air quality data along with the **Air Quality Index (AQI)** where you live and in other locations around the U.S. and worldwide.

AirNow Homepage

Type in a **Zip Code**, **City**, or **State** to get air quality data anywhere, anytime!



Chicago,
IL 60612
Air Quality
Index,
June 22,
2025,
21:00

<https://www.airnow.gov/>

AQ 101: Understanding the AirNow Graph

❖ **AQI for Chicago on June 22, 2025, at 21:00**

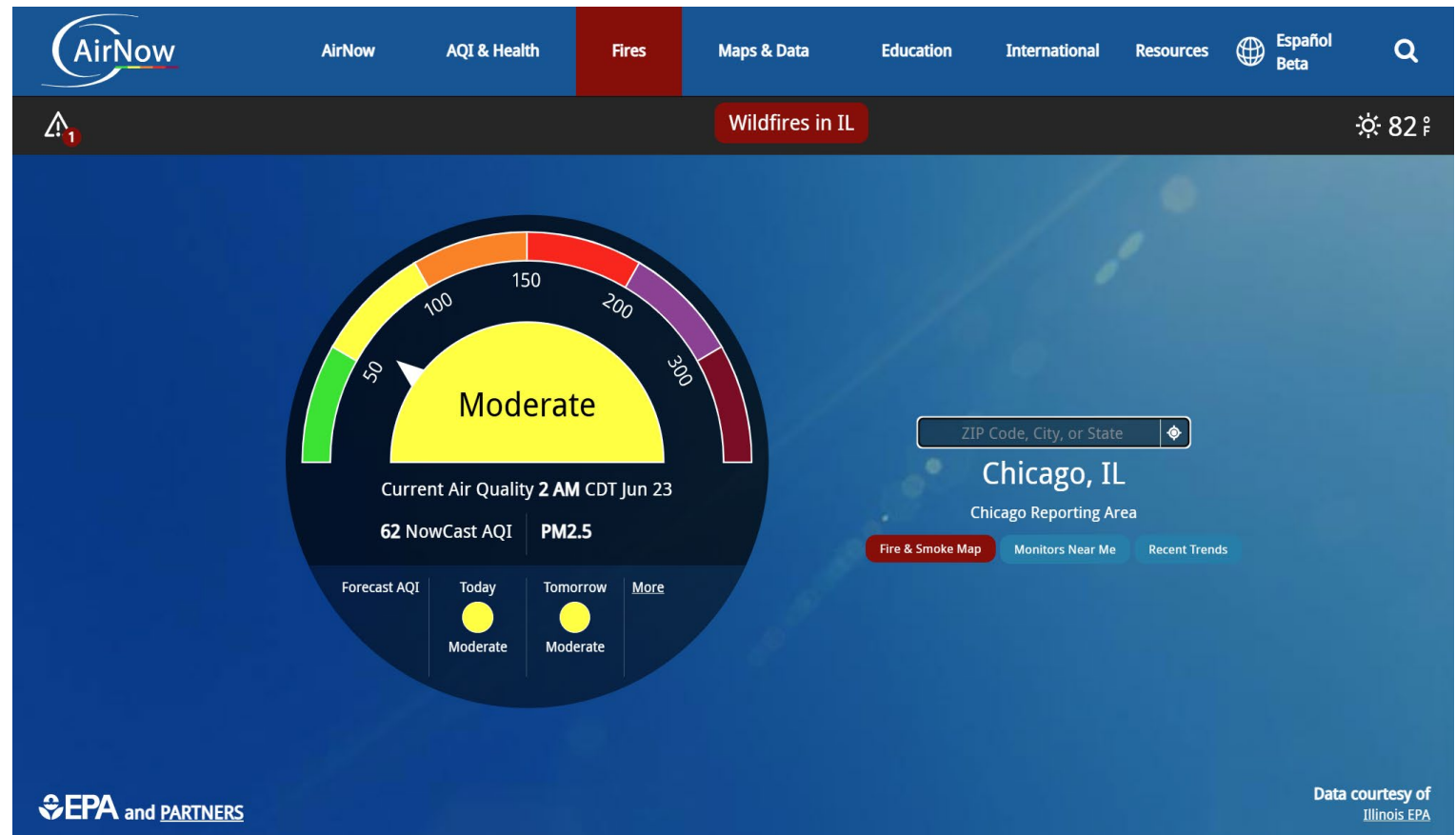
Color = Yellow

Level of Concern = Moderate

AQI Value = 62

Description = Air quality is acceptable. However, there may be a risk for some people, particularly those who are unusually sensitive to air pollution

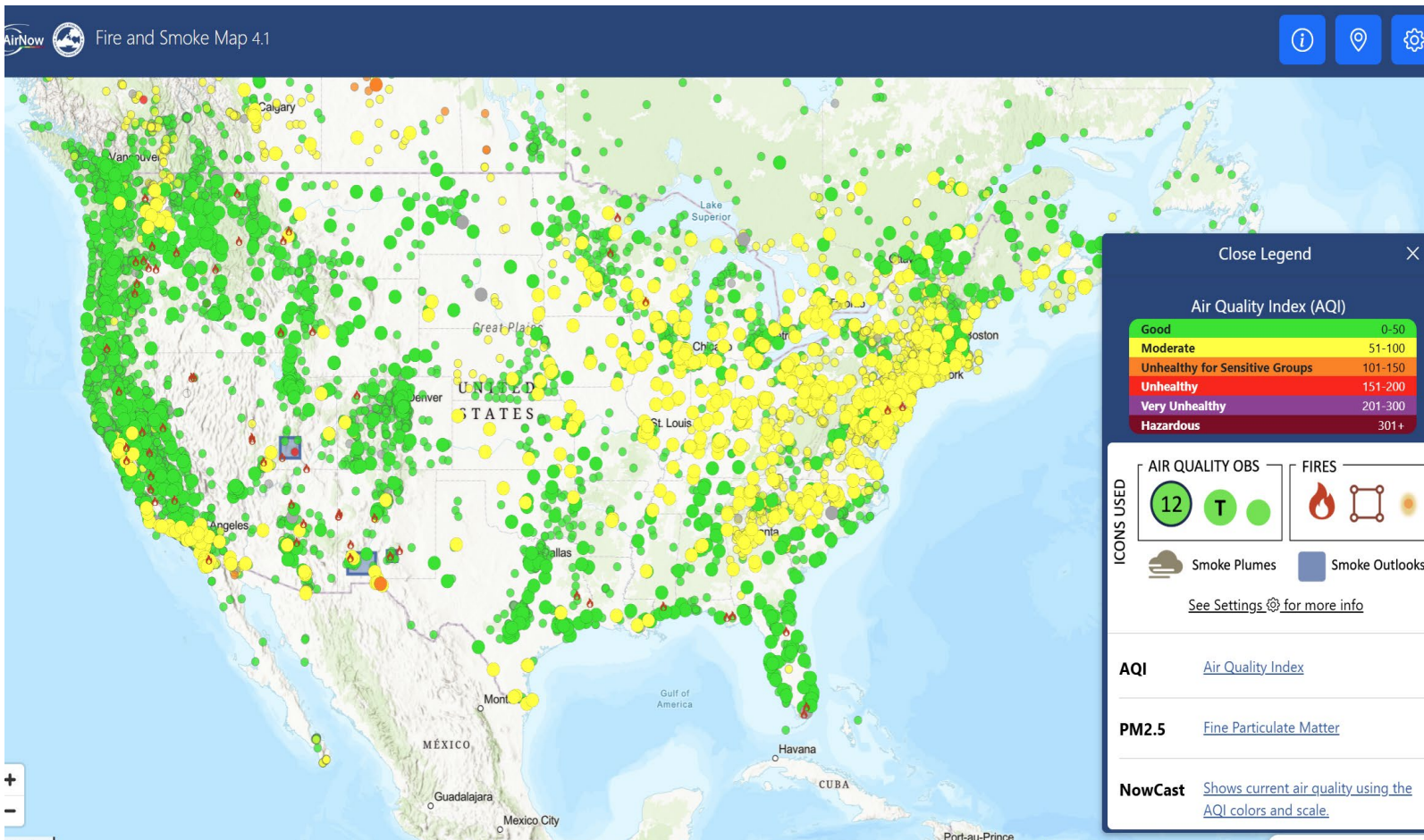
Air quality data and AQI shown is for PM2.5



Information updated every hour with a rolling average!

AQ 101: AirNow – Current Fire Conditions

AirNow provides information Fire and Smoke Map for the U.S., showing current wildfires, wildfire incidents, air quality data, as well as resources on how to protect your health (<https://fire.airnow.gov/>).



Fire and smoke data is provided by the U.S. Forest Service and the National Oceanic & Atmospheric Administration (NOAA).

❖ **Colored circles (in green/yellow in this map) represent current AQI at USEPA Air Monitoring Stations across the U.S.**

❖ **Red Dots - Active Fires and Wildfire Incidents**

AQ 101: Methods for Determination of Ambient Air Concentrations of Air Pollutants (Exposure Assessment)



Ambient Air Concentration

Measurements-based

+

Modeling-based

Predictive Modeling using Machine Learning Methods

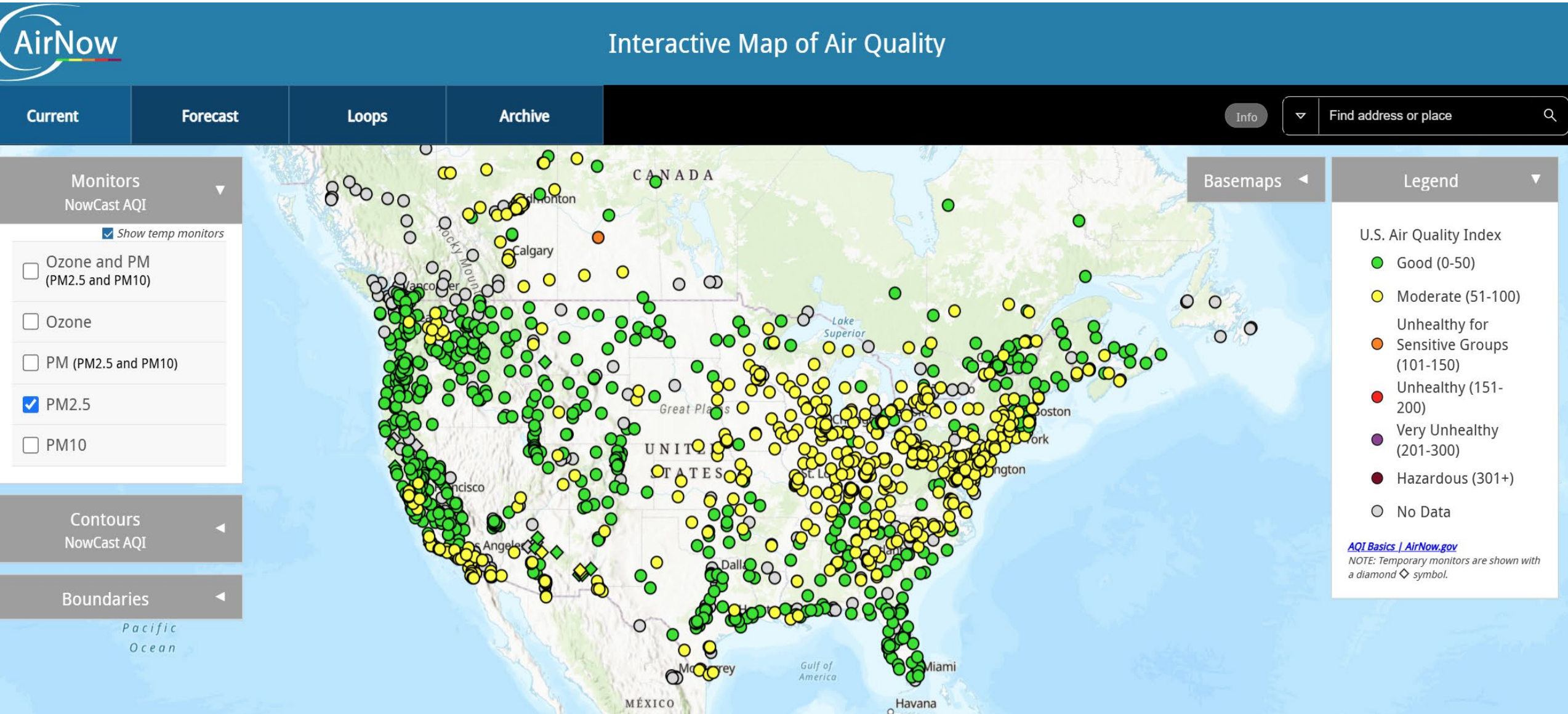
Air Dispersion Modeling

Geospatial (GIS) Modeling

EPA Methods
**Federal Reference Monitors (FRMs)
** Federal Equivalent Monitors (FEMs)

**Near Reference Air Monitors
** Sensors for Area (Community) Monitoring
**Sensors for Personal Exposure Monitoring

AQ 101: Where Does the Air Quality Data Come From?



Source: <https://gispub.epa.gov/airnow/>

AQ 101: Who Monitors Air Quality?

- ❖ **State, local, and/or tribal agencies** are responsible for monitoring air pollutants
- ❖ Monitoring data is sent to the U.S. EPA, data subjected to thorough data cleaning and quality assurance and quality control (QA/QC), statistical data analysis per EPA protocols.
- ❖ Data displayed on the AirNow website and verified data stored in the Air Quality System Database

Wait – I thought the U.S. EPA monitors air quality everywhere?

- ❖ EPA delegates authority to monitor air pollution to states (sometimes states delegates authority to locals)
- ❖ EPA's role is to...
 - ❖ Set the regulations such as the NAAQS levels
 - ❖ Assess compliance with criteria pollutant NAAQSs
 - ❖ Provide guidelines on how to monitor air pollutants using FRMs/FEMs
 - ❖ Ensure states comply with air quality regulations

Example of a Regulatory Monitoring Site on a Trailer



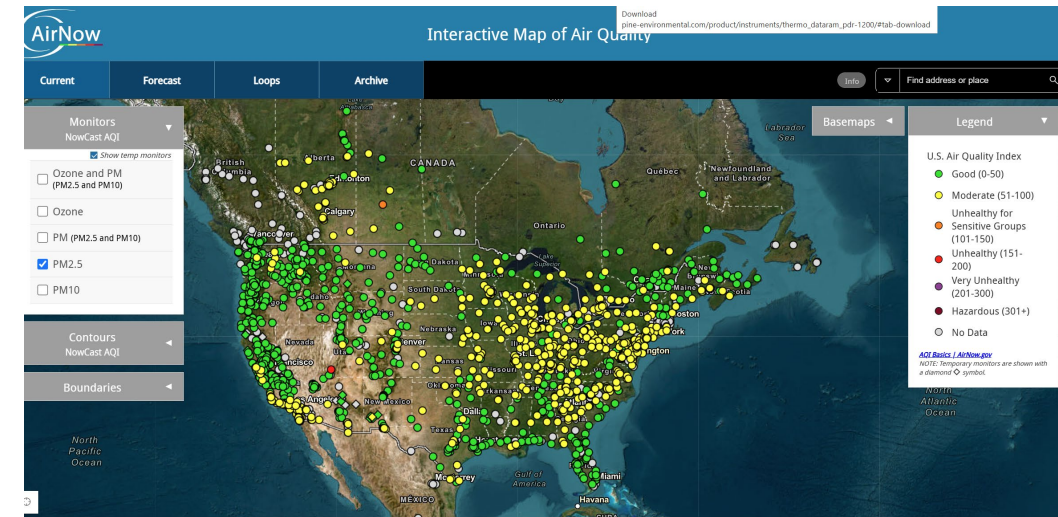
Gaseous pollutant measurement instruments



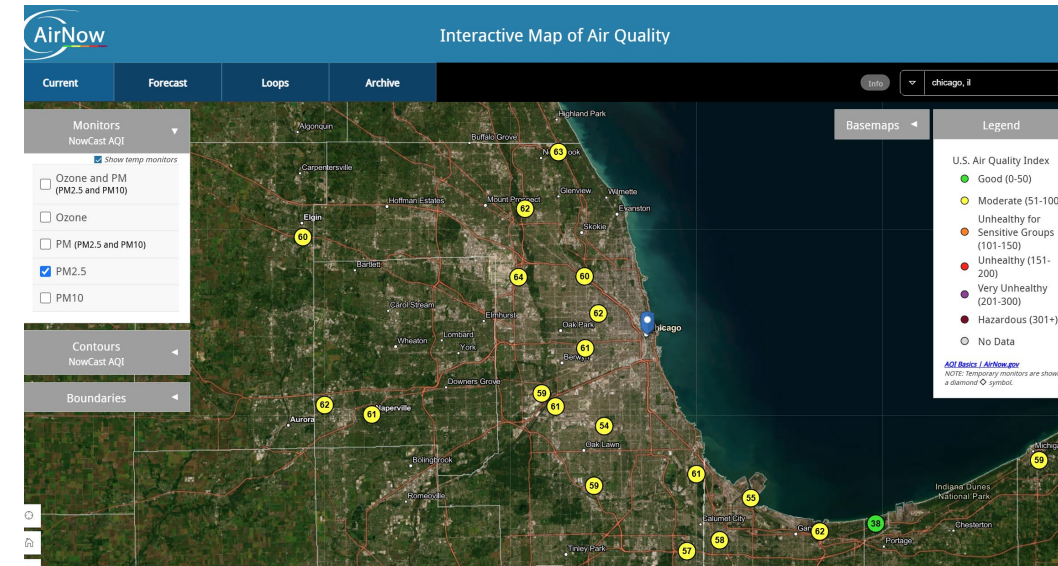
AQ 101: Regulatory Monitoring with FRMs/FEMs

- ❖ Uses EPA-certified equipment: Meet strict operating and performance requirements using higher-cost and higher quality instrumentation
 - ❖ **Federal Reference Method (FRM)**
 - ❖ **Federal Equivalent Method (FEM)**
- ❖ Follows precise operating procedures defined in Code of Federal Regulations
 - ❖ Sited according to very intentional and specific criteria
 - ❖ Undergo strict and routine maintenance
 - ❖ Data evaluated according to specific protocols
- ❖ Performed by state, local and tribal air quality agencies
- ❖ Network does not cover all communities
- ❖ **Data used to assess compliance with health-based NAAQS and for air quality control and management and health effects research**

U.S Map of EPA FRMs/FEMs for PM2.5 and AQI



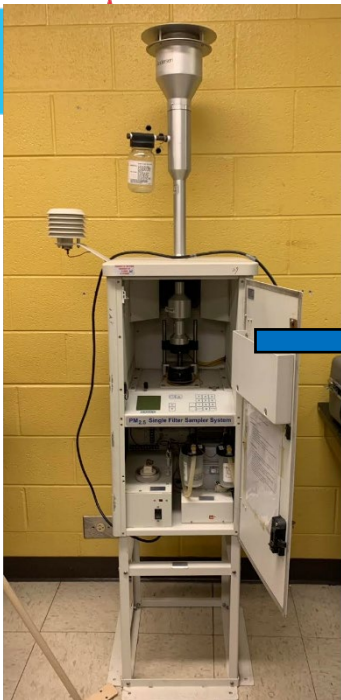
Chicago Area Map of EPA FRMs/FEMs for PM2.5 and AQI



AQ 101: Why are FRMs/FEMs used for regulatory decision-making?

Flow Rate=Velocity * Surface Area

Momentum = Mass * Velocity



EPA PM2.5 FRM



EPA PM2.5 WINS (well-type impactor ninety-six) Impactor

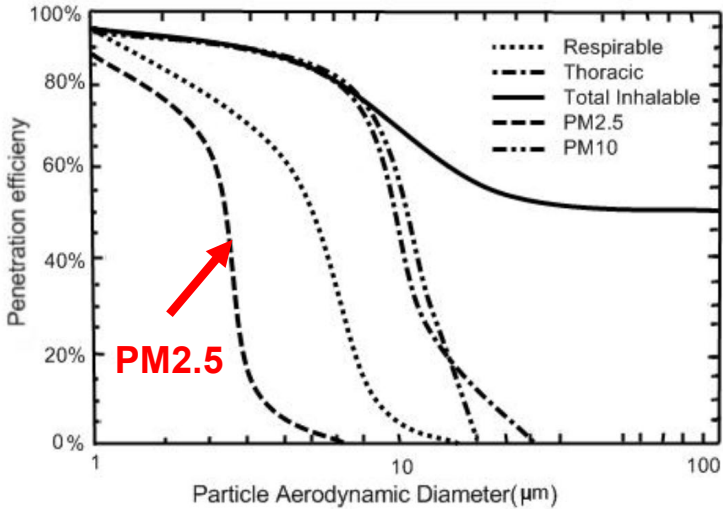
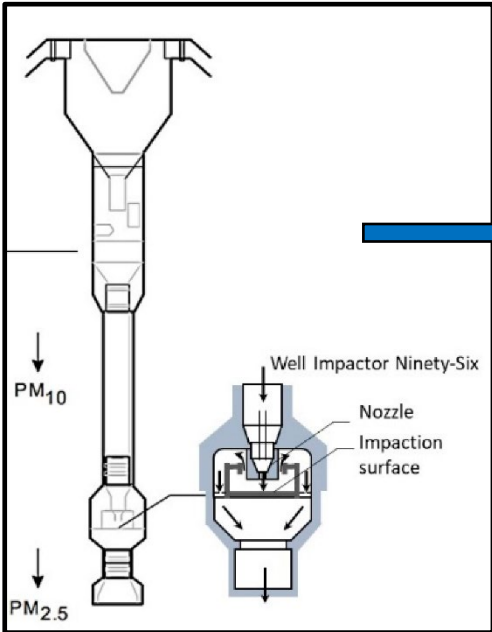
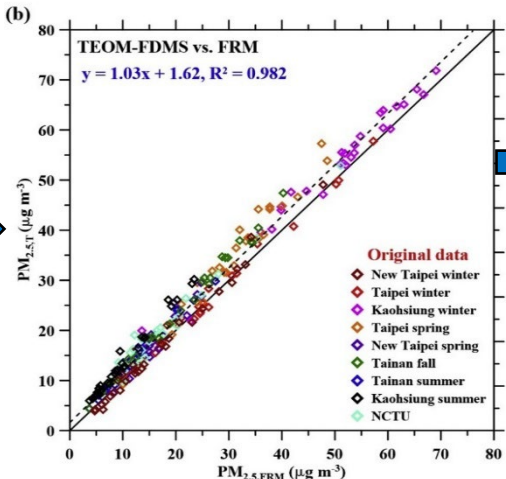


Figure 4-2. Modeled lung deposition as a function of particle size.

EPA PM2.5 WINS Impactor-Biologically relevant sampling-Instrument measures PM2.5 fraction deposited in lung as a measure of PM2.5 exposure in humans

FEMs are calibrated against FRMs in field and laboratory conditions by the EPA



AQI=(FRM/FEM Concentration/NAAQS)*100

Measure of public health significance of measured FRM/FEM air concentrations

Developed from scientific health effects studies describing dose-response relationships for criteria air pollutants

AQ 101: Newer Ways to Collect Air Quality Data – Air Sensors!

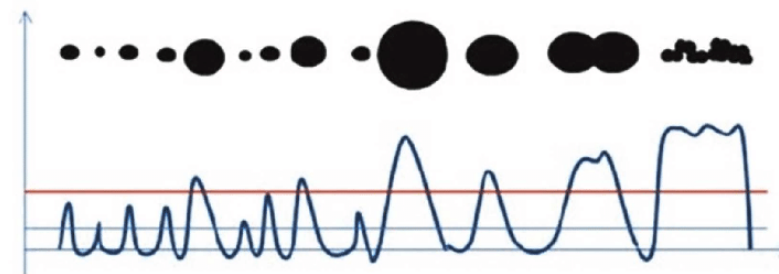
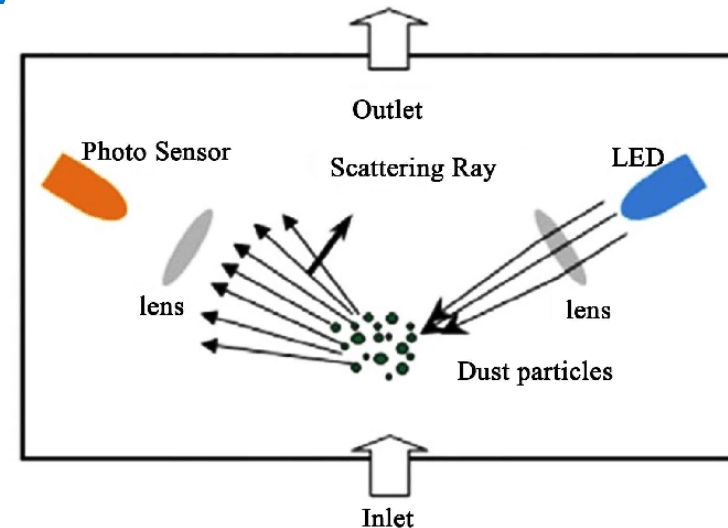
- ❖ **Air sensors** are devices that can measure a variety of air pollutants (both particulates and gaseous pollutants)
- ❖ Sensors are often smaller, lower in cost, portable, less accurate compared to the instruments (FRMs/FEMs) used for EPA's regulatory monitoring
- ❖ Sensors are being used in many ways....
 - ❖ Collecting local or neighborhood air quality measurements to assess spatial variation in air quality
 - ❖ Supporting community/citizen education and environmental awareness efforts
 - ❖ Conducting scientific research
 - ❖ Conducting community/citizen science projects
- ❖ The advancement on sensor technology enabled real-time measurement of air pollutants in the last decade. Data quality from sensors varies and knowledge on reliability and accuracy of sensor are under development for many sensors.
FRMs/FEMs, part of the EPA's regulatory monitoring network, are the "gold standard" for accuracy and are used to ensure compliance with NAAQS. Air quality sensors, on the other hand, are, useful tools for developing supplemental air quality information about air quality in communities, especially in areas not covered by the regulatory network.



Sensors **cannot** be used for regulatory monitoring, compliance assessment with NAAQS or enforcement actions!

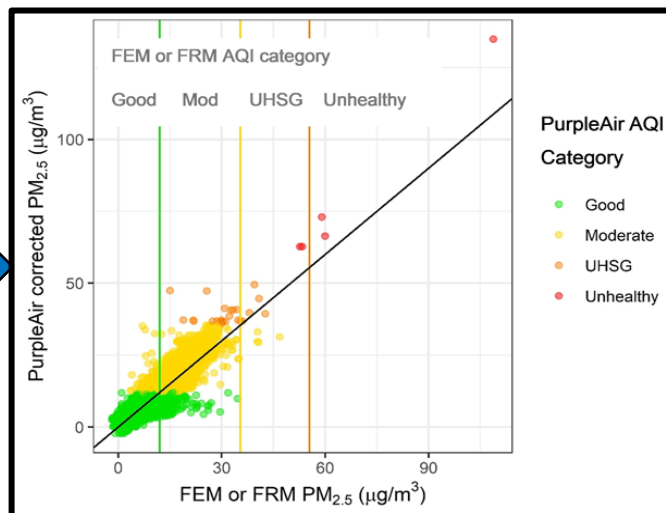
AQ 101: Why sensor data cannot be used for regulatory decision-making?

- ❖ Sensor measurements do not mimic human exposures, unlike FRMs. Sensors provide relative concentration of PM_{2.5} in ambient air in a geographic space (high concentration locations vs. low concentration locations). The measurements have no biological significance on their own.
- ❖ PM_{2.5} sensors commonly use the light scattering method. They work by drawing in air and shining a light source (often a laser) through it. As PM_{2.5} particles pass through the light source, they scatter the light in various directions. A detector measures the intensity and pattern of the scattered light, which is then used to determine the size and number of particles, and subsequently, the PM_{2.5} number concentration (number of particles/cm³), which is converted to PM_{2.5} mass concentration (mg/m³).



Collocation Study:

Sensors are calibrated against FRMs/FEMs at EPA air monitoring sites across four seasons to be able to indirectly assess public health significance of sensor data collected



$$AQI = (\text{Corrected Sensor PM}_{2.5} \text{ Concentration} / \text{NAAQS}) * 100$$

Measure of public health significance of corrected sensor concentrations measured in ambient air

Developed from scientific health effects studies describing dose-response relationships for criteria air pollutants

Source: <https://amt.copernicus.org/articles/14/4617/2021/>

AQ 101: SA2 QA/QC: Sensor Performance Assessment at a IEPA Air Monitoring Site

Collocation Study at IEPA Northbrook, IL Air Monitoring Site

MO: MetOne PM Neighborhood Monitor

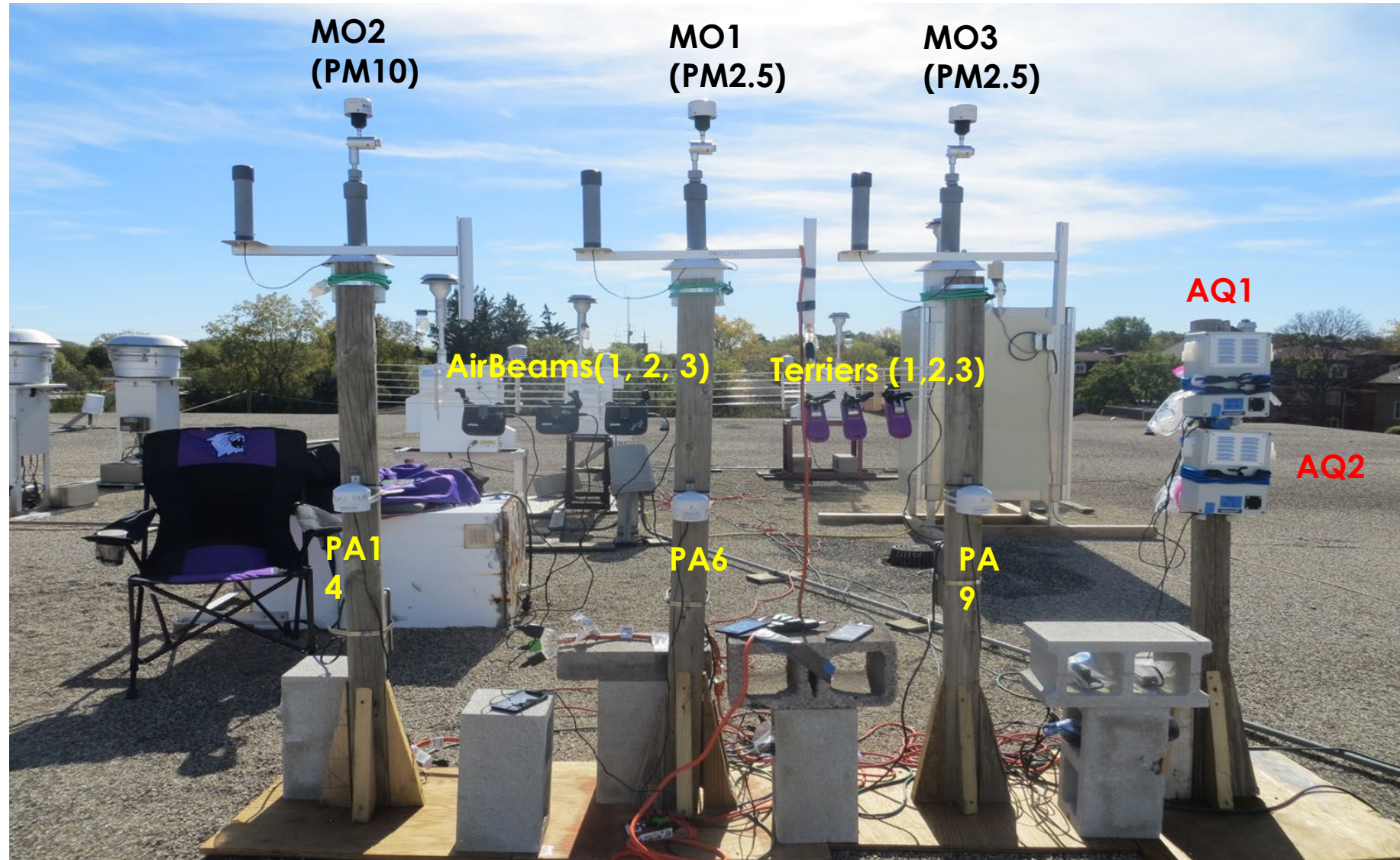
PA: PurpleAir PM sensor

AirBeam: PM Sensor

Terrier: Terrier NO/CO/CO₂ Sensor

AQ: AeroQual NO₂/O₃ Sensor

Note: Experiment was conducted from October 6 to December 8 (1512 hours) with a sub-experiment with AirBeam/Terrier sensors lasting for 84 hours.



This work was performed by Dr. Erdal and her Ph.D. student, Dr. Saisattha Noomnuan.



AQ 101: Air Sensor Strengths- Some correlate reasonably well with FRMs/FEMs to be used for...

- ❖ **Increasing** the capability to provide continuous, real-time data on pollutant levels – increases the ability to take prompt action for reducing exposures or timely public health messaging – can serve as an early warning system. This can lead to reduced health risks associated with air pollution,
- ❖ **Helping** individuals make decisions about the timing and location of daily activities like commuting, staying indoors on days with poor air quality, and exercise, leading to exposure and health risk reduction through personal behavioral changes.
- ❖ **Supplementing** existing monitoring networks to increase spatial coverage and fill knowledge gaps. Granular hyperlocal data helps researchers and policymakers to understand spatial and temporal trends in air quality with a better ability to develop hypothesis on relationships between air pollution sources and exposures at the community level – leading to more informed decision making for air quality control and management and public health improvement
- ❖ **Measuring** smaller scale variations in spatial concentration or determining how air pollution sources may be affecting a community and **Providing** a means to learn about air quality issues potentially impacting a community, sources leading to air pollution, and variations in air quality in different areas of the community – a more comprehensive assessment of air quality at community level
- ❖ **Improving** the ability to establish a better air monitoring infrastructure by purchasing and operating a greater number of sensors, resulting in a denser air monitoring network, and yielding improved understanding of spatial and temporal patterns in a community Air sensors are often more affordable than EPA FRMs and FEMs and requires less manpower to operate and maintain as compared to FRMs/FEMs.
- ❖ **Empowering** communities and public in general with air quality information, supporting community organization outreach, education, advocacy for their constituents. The data are especially valuable for communities disproportionately impacted/burdened by air pollution.
- ❖ **Enhancing** scientific research infrastructure to study effects on air pollution on health, economy, ecosystem, and well-being. Air quality sensor data can support research studies to investigate the health effects of air pollution, evaluate the impact of different interventions, and develop new air quality management strategies. For example, researchers can analyze sensor data to study the short-term and long-term effects of air pollution on pediatric health.

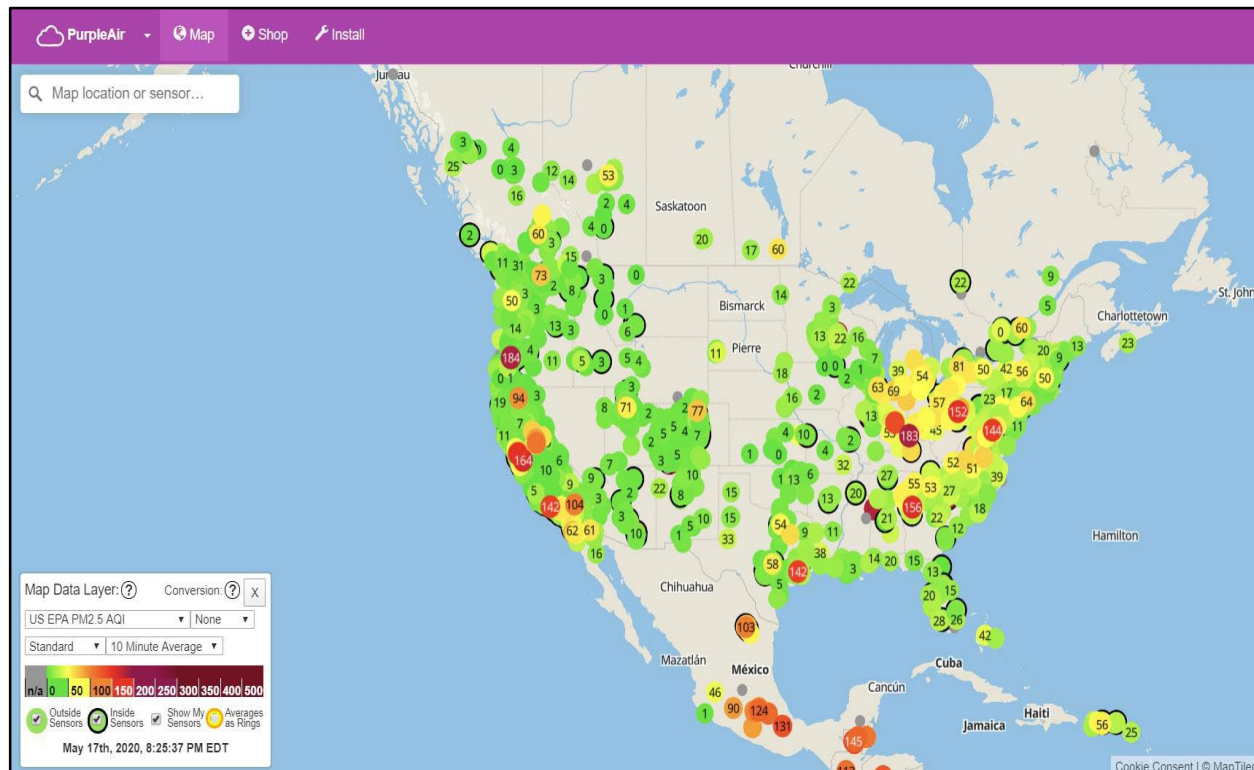
AQ 101: Air Sensor Weaknesses



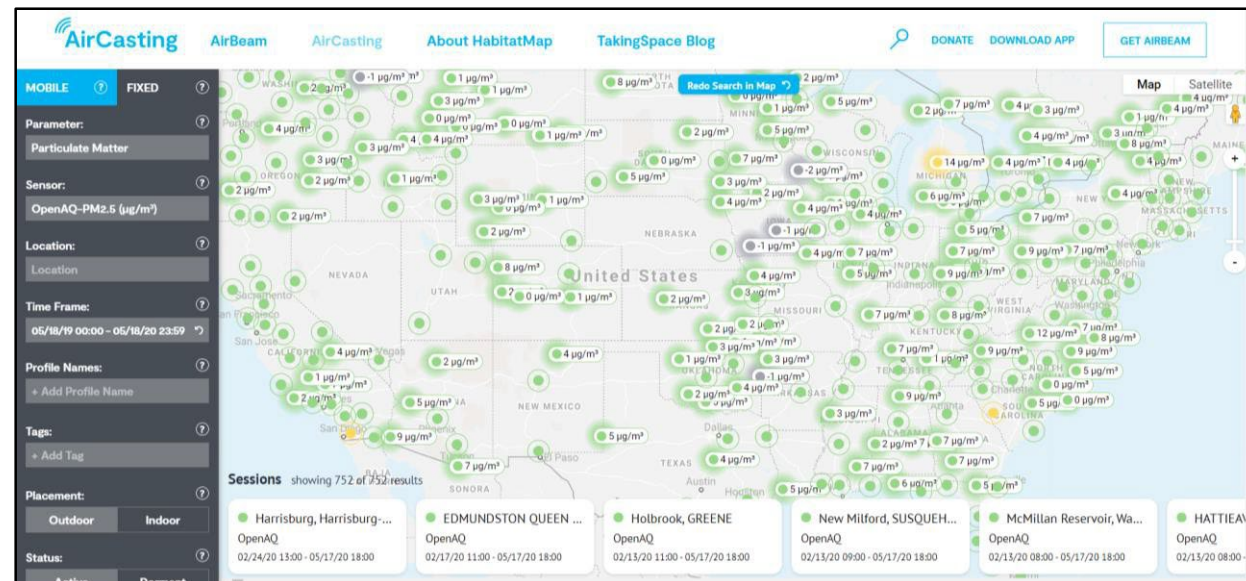
- ❖ Involves a wide range of devices: high to lower quality with a wide range on sensor performance and reliability
- ❖ **Air sensors may not always be accurate (yielding high concentration data when the concentrations are not high or vice versa) and require calibration to produce more precise data. The sensor performance must be tested against the EPA FRM/FEM monitor in field conditions and calibration equation spanning all weather conditions and concentration ranges must be developed. This is time-consuming and requires, at least, a year of testing (across four seasons) the sensor and operating next to EPA FRMs/FEMs at an EPA monitoring site (see Slide 42). This is called a collocation study in which performance of a device (e.g., an air sensor) is tested by installing and operating near a reference instrument (e.g., FEPA FRM/FEM). Based on the calibration equation developed in the collocation study, the sensor data may need to be adjusted to improve accuracy. The adjusted sensor data is called corrected sensor data. Collocation and correction steps follow one another.**
- ❖ Can be undertaken by anyone: agencies, community groups, researchers, businesses, or citizens
- ❖ Sensor performance is influenced by temperature and relative humidity. Low and high temperatures, as well as high humidity levels, can cause inaccurate readings.
- ❖ **No established standard protocols or certification programs for the accuracy, precision, calibration, siting, operation, maintenance, measurement methods, instrument audits and instrument QA/QC, data QA/QC – Thus, quality of data collected is of concern – the data cannot withstand legal scrutiny. There are no federal standards or certification This variability makes it difficult to compare data collected at different cities or collected by different entities.**
- ❖ Technical expertise needed for data interpretation. In addition, special technical expertise needed for data management tasks, such as data handling, data QA/QC, processing, and analyzing the large amount of data generated by a dense air monitoring network
- ❖ **Cannot be used for regulatory decision-making or compliance or enforcement purposes (e.g., to assess whether NAAQSs are legally met in a given geographic area, or whether a source/facility is in compliance with its operating permit conditions, etc.)**

AQ 101: Air Sensors – Some Data is Available for the Public

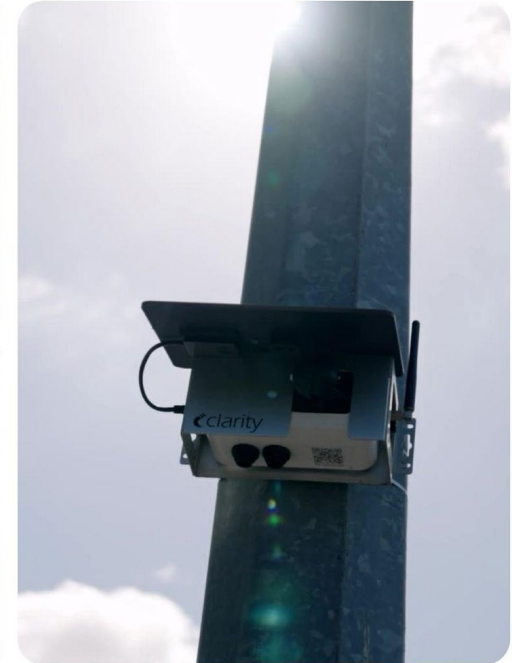
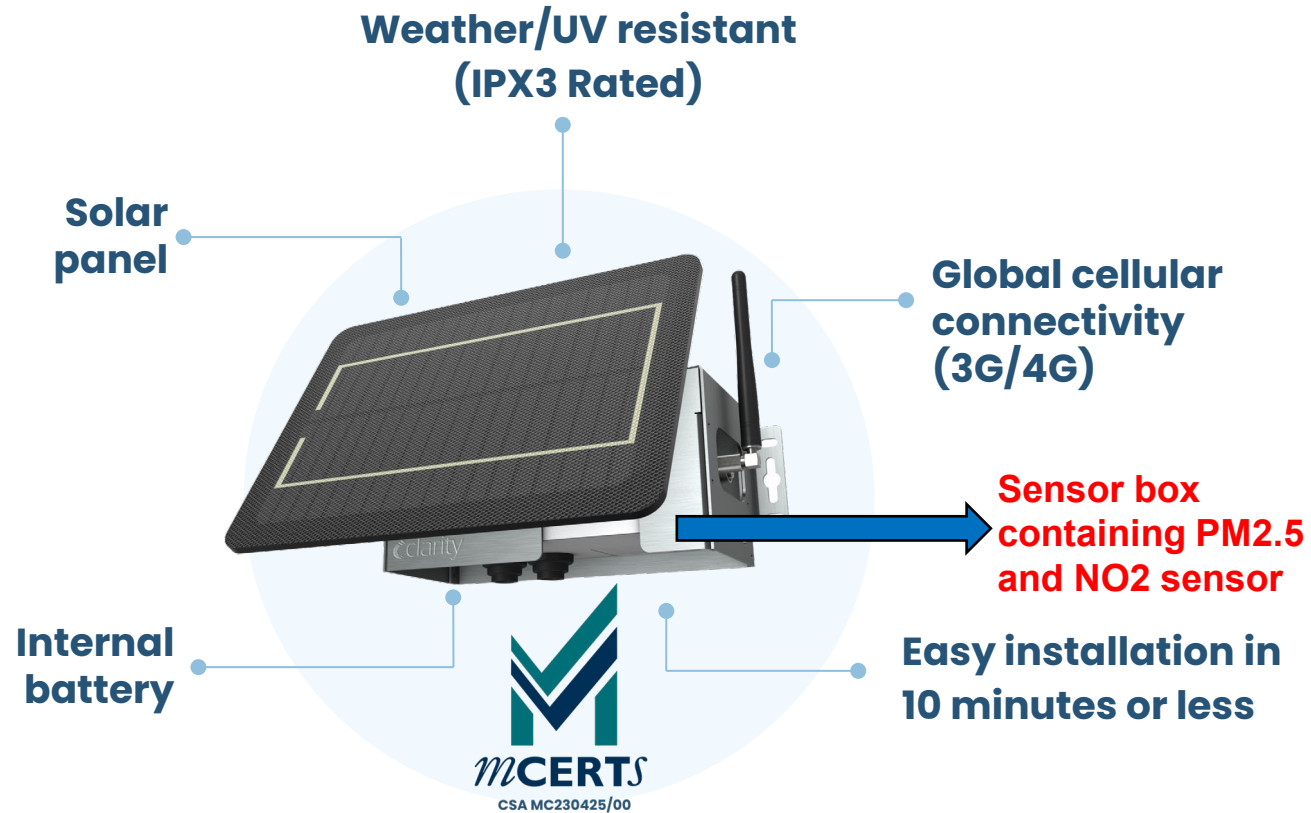
PurpleAir Map (PurpleAir Sensor) – Fee for data download
<https://www2.purpleair.com/>



AirCasting Map (AirBeam sensor)
http://aircasting.habitatmap.org/mobile_map



AQ 101: Chicago's Sensor-based AQ Network: Clarity Node-S

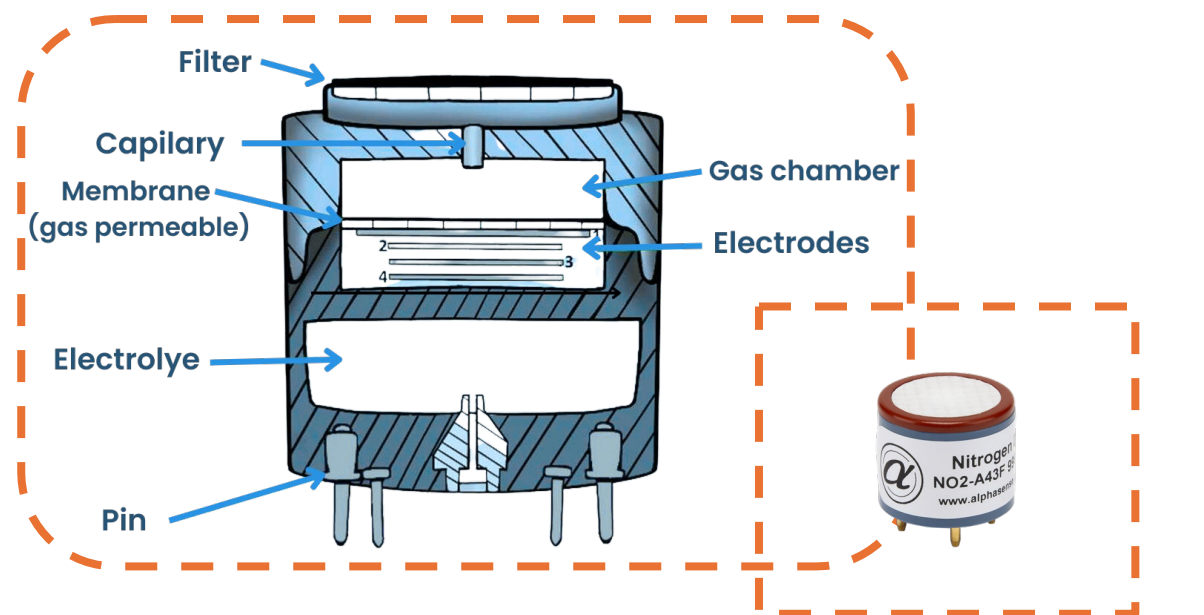
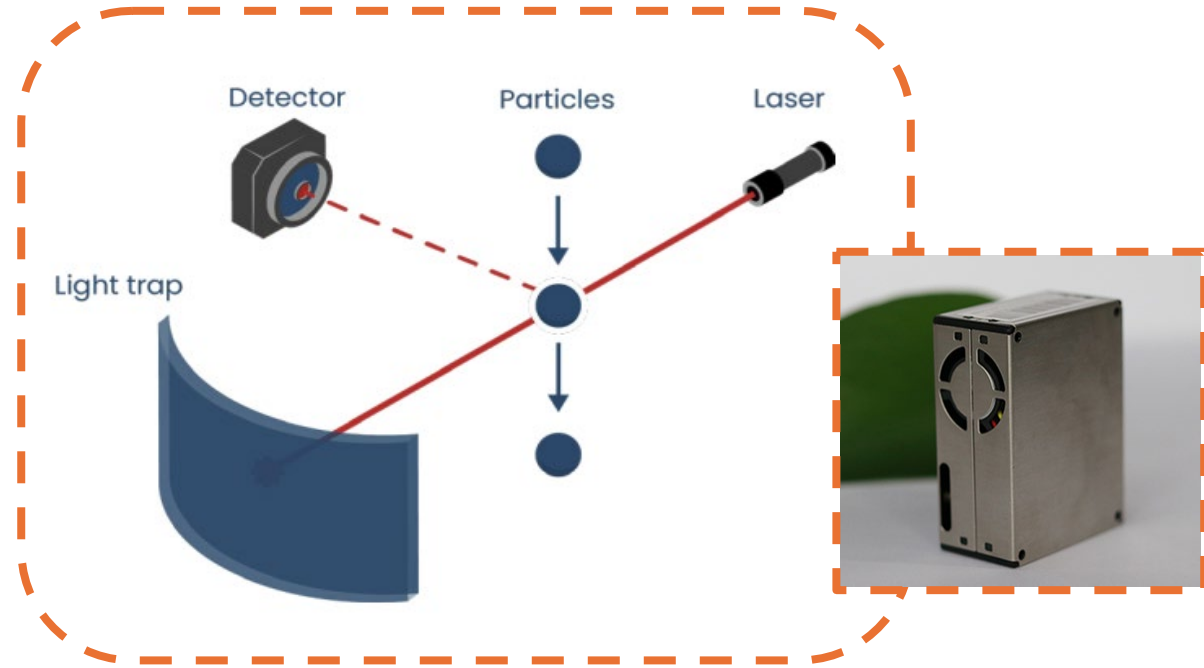


AQ 101: Clarity Node-S:

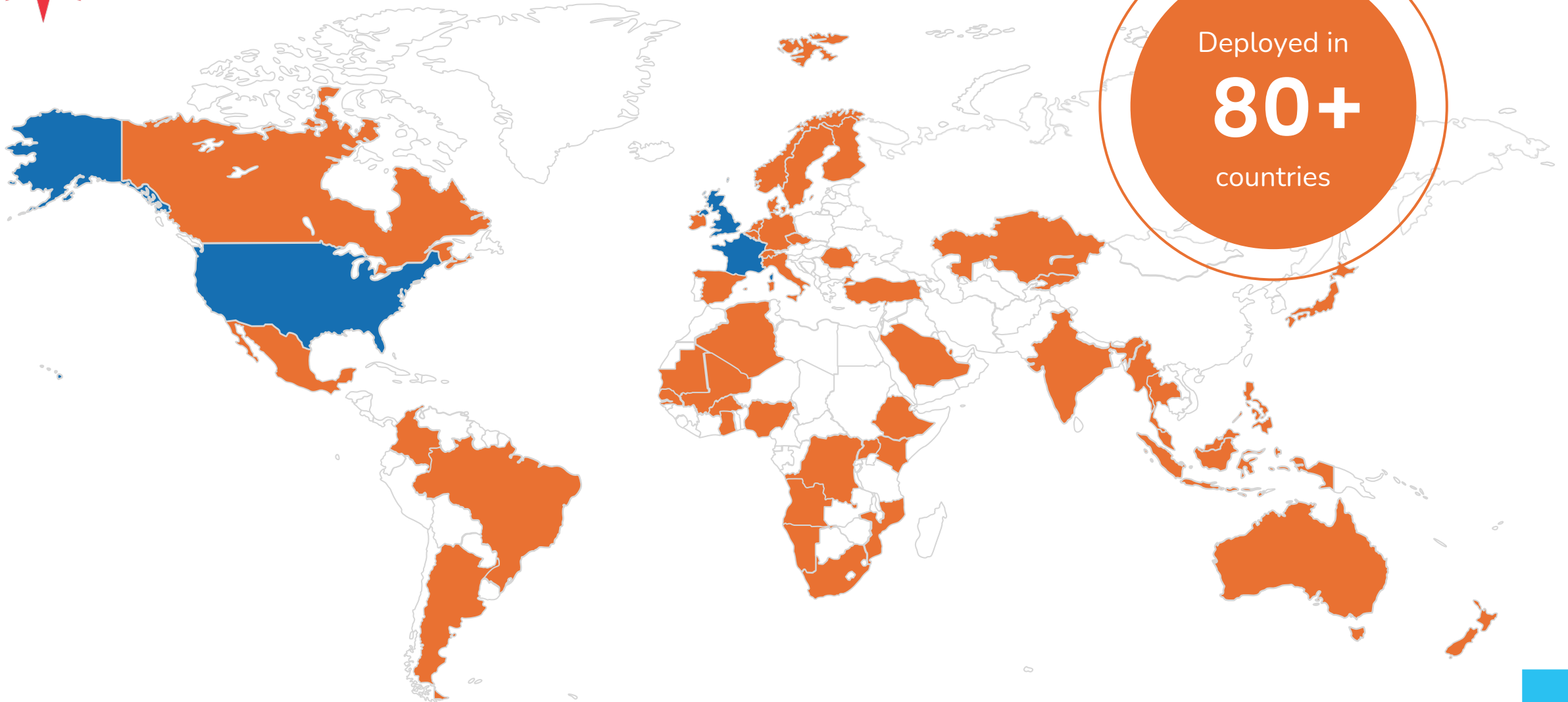
Sensor Specifics for PM2.5 and NO₂

PM2.5 Measurement Method: The OPC included in the Clarity Node-S is manufactured by Plantower (PMS6003) — this is a more robust, dual laser version of the Plantower PMS5003 sensor used in the Purple Air PA-II. Sensor uses laser light scattering principle to measure value of dust particles suspended in the air.

NO₂ Measurement Method: The electrochemical cell sensor included in the Clarity Node-S is manufactured by Alphasense (NO₂-A43F) — the best performing low-cost NO₂ sensor on the market. Alphasense that operate in an amperometric mode. That is, they generate a current that is linearly proportional to the fractional volume of the toxic gas in the environment (see <https://docs.iscape.smartcitizen.me/Components/Gas%20Pro%20Sensor%20Board/Electrochemical%20Sensors/>)



AQ 101: Clarity Sensor Global Network



Deployed in

80+

countries

AQ 101: Ways to Get Air Quality Information in the U.S.

❖ AirNow

- Visit the AirNow website at <https://www.airnow.gov/>
- AirNow App (available on iPhone and Android)



❖ AirNow EnviroFlash

- System that sends you e-mails about your daily air quality forecast
- Visit <http://www.enviroflash.info/> to sign up!

❖ AQI is also reported in...

- Weather forecasts on local news, radio stations, and newspapers
- State, local, or tribal air quality agency websites



❖ EPA's Air Quality System (AQS)

- The Air Quality System (AQS) contains ambient air pollution data collected by EPA, state, local, and tribal control agencies from over thousands of monitors across the U.S. AQS also contains meteorological data, descriptive information about each monitoring station (including its geographic location and its operator), and data quality assurance/quality control information. <https://www.epa.gov/aqs>. **There is about 4-6 months lag time between data collection and data availability on the AQS site because the data collected goes through rigorous data quality assurance and quality control (QA/QC), data evaluation, and data analysis prior to public release.**

❖ PurpleAir Map

- PurpleAir PM2.5 data can be downloaded for a fee (unless one has points accumulated from PurpleAir, Inc.). The quality of PurpleAir data is of question because there is no standard protocol is implemented in the sensor installation and operation by those who install these sensors (<https://community.purpleair.com/t/purpleair-data-download-tool/3787>)

❖ American Lung Association's State of the Air report

- This report analyzes air quality data from official monitors and provides grades and rankings for different locations across the country, including Chicago, IL (<https://www.lung.org/research/sota>)

❖ NASA Satellite Data

- NASA uses data from its Earth-observing satellites to monitor and study air quality. These satellites carry instruments that measure various pollutants like nitrogen dioxide, ozone, and particulate matter and atmospheric conditions (<https://airquality.gsfc.nasa.gov/>)

Additional Resources

- ❖ EPA Air Sensor Toolbox (<https://www.epa.gov/air-sensor-toolbox>)
- ❖ EPA Managing Air Quality: Ambient Air Monitoring Program (<https://www.epa.gov/air-quality-management-process/managing-air-quality-ambient-air-monitoring>)
- ❖ EPA, Interactive Map of Air Quality Monitors,
- ❖ <https://www.epa.gov/outdoor-air-quality-data/interactive-map-air-quality-monitors>
- ❖ EPA, National Air Quality: Status and Trends of Key Air Pollutants, <https://www.epa.gov/air-trends>
- ❖ EPA Air Sensor Evaluations Conducted by Other Organizations, <https://www.epa.gov/air-sensor-toolbox/air-sensor-evaluations-conducted-other-organizations>
- ❖ South Coast Air Quality Management District, Air Sensor Performance Testing Center, <https://aqmx.org/resources/air-quality-sensor-performance-evaluation-center-aq-spec>
- ❖ PurpleAir, Chicago, Real Time Map link, <https://map.purpleair.com/air-quality-standards-us-epa-aqi?opt=%2F1%2Fb%2Fm%2Fi%2Fip%2Fa0%2Fp1800%2FcC5#10.14/41.8399/-87.6831>
- ❖ AirBeam, Chicago, Real Time Map Link, https://aircasting.org/?_gl=1*14z2img*_gcl_au*MTM2NzQzODIyOS4xNzUwNjcwNjk5*_ga*MTEwODY4Njg4MC4xNzUwNjcwNjk5*_ga_8MP8N81Y25*czE3NTA2NzA2OTkkbzEkZzAkdDE3NTA2NzA2OTkkajYwJGwwJGg1NzlyOTI1ODA.&sessionId=&streamId=&isActive=true&sessionType=fixed&boundEast=-87.22725982224334&boundNorth=42.04330086687738&boundSouth=41.62399410585339&boundWest=-88.23649817775664&thresholdMin=0&thresholdLow=9&thresholdMiddle=35&thresholdHigh=55&thresholdMax=150¤tCenter=%7B%22lat%22%3A41.833990851026506%2C%22lng%22%3A-87.73187899999999%7D¤tZoom=11.041272643394183

Summary

We learned about....

- ❖ **Definitions** of air pollution and air quality
- ❖ **History** of air pollution
- ❖ **Regulation** of air pollution in the U.S.
- ❖ **Common pollutants** in the air
- ❖ **Sources** of air pollution
- ❖ **Health effects** of air pollution
- ❖ **Measurement methods** for determining ambient air concentrations of pollutants (EPA's FRMs/FEMs vs. Sensors)
- ❖ **Air Sensors** - newer ways to collect air quality measurements (strengths and weaknesses of sensors)
- ❖ **Common questions**
 - ❖ Why and when air pollution can be higher and lower?
 - ❖ How to tell what the air quality is and where to find information?
 - ❖ How are air pollutants measured and where to find the data?

Additional References

In addition to the references cited in each slide, the following references were used:

- ❖ U.S. Environmental Protection Agency, Rachelle Duvall*, Andrea L. Clements*, and Ryder Freed+, Air Quality 101, A Background on Air Pollution, *Office of Research and Development (Research Triangle Park, NC), +Region 9 (San Francisco, CA), May 2020.
- ❖ Dr. Erdal's EOHS 421 and EOHS 494 Courses PowerPoint Presentations and EPA slides from the Air Pollution Training Institute Workshops
- ❖ EPA, Air Pollution Training Institute, APTI 435: Atmospheric Sampling , Instructor's Manual, Manual, EPA 450-2-80-006, Environmental Research Center, Research Triangle Park, NC, February 1980.

Note: The presenters build this presentation content by augmenting, updating content into the EPA presentation by Duvall et al. (2020) listed above, along with adding new content per their professional expertise.